Smart Home

CST2565 Project Report

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CST2565 Assignment 2 – Group Project

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Individual Contribution to Group Project	Peer Assessment: Everyone worked outstanding, Equal contribution.
Outstanding	The contribution was significantly OVER AND ABOVE the performance of any other group members; hence ONLY ONE member per group can have this ranking
Good	The contribution was valuable and significant either in content or in underpinning the group as a whole.
Average	The contribution was average. No more or less than could reasonably be expected.
Poor	The contribution was minimal OR the member did not contribute at all.

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Rabia Bint Abdul Ahad Abdul Rahim	M01035049	Equal Contribution	• Led the design of the **starter homes (8×2)**, focusing on cost-effective and scalable smart solutions • Designed the system architecture, including device, edge, cloud, and application layers • Developed the mobile application interface, including user flow, control features, and real-time alert integration • Conducted smart device research and selection based on compatibility, cost, and performance • Performed cost analysis, budgeting, and financial evaluation including ROI and payback period • Contributed to energy efficiency planning through solar integration and smart optimisation strategies	

			• Supported security design, including encryption, access control, and compliance with ISO standards and UAE regulations • Participated in system integration discussions to ensure seamless communication between devices • Contributed to risk management by identifying system vulnerabilities and proposing mitigation strategies • Assisted in project management tools including WBS, Gantt chart, and critical path analysis • Contributed to report writing, formatting, and overall project documentation • Participated in presentation design and preparation • Actively contributed to brainstorming, idea development, and design decision-making	
Raghad Khalid Barmem	M01039213	Equal Contribution	• Led design of the **family homes**, focusing on safety, automation, and convenience • Designed the system architecture, including device, edge, cloud, and application layers • Developed the mobile application interface, including user flow, control features, and real-time alert integration • Conducted smart device research and selection based on compatibility, cost, and performance • Performed cost analysis, budgeting, and financial evaluation including ROI and payback period • Contributed to energy efficiency planning through solar integration and smart optimisation strategies • Supported security design, including encryption, access control, and compliance with ISO standards and UAE regulations • Participated in system integration discussions to ensure seamless communication between devices • Contributed to risk management by identifying system vulnerabilities and proposing mitigation strategies • Assisted in project management tools including WBS, Gantt chart, and critical path analysis • Contributed to report writing, formatting, and overall project documentation • Participated in presentation design and preparation • Actively contributed to brainstorming, idea development, and design decision-making	

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1) INTRODUCTION

Smart technologies refer to modern systems and devices that are designed to operate in an intelligent, efficient, and adaptive manner. These technologies extend traditional automated systems by integrating components such as sensors, actuators, and embedded controllers, enabling systems to respond dynamically to environmental changes and user requirements.

Smart technologies are a fundamental part of the Internet of Things (IoT), where interconnected devices communicate through networks to collect, process, and exchange data (Atzori, Iera and Morabito, 2010). Through this connectivity, systems can analyse real-time information and automatically adjust operations to improve performance, efficiency, and user experience.

In residential environments, smart home technologies combine hardware and software to enable automation, remote control, and intelligent decision-making. These systems often incorporate artificial intelligence (AI), data analytics, and machine learning techniques to enhance functionality, such as energy optimisation, security monitoring, and user behaviour prediction The adoption of smart home technologies has increased significantly due to their ability to reduce energy consumption, improve security, and support independent living, particularly for elderly or vulnerable individuals (Wilson, Hargreaves and Hauxwell-Baldwin, 2015). As a result, smart homes are becoming an essential component of modern residential design, offering both practical and economic benefits.

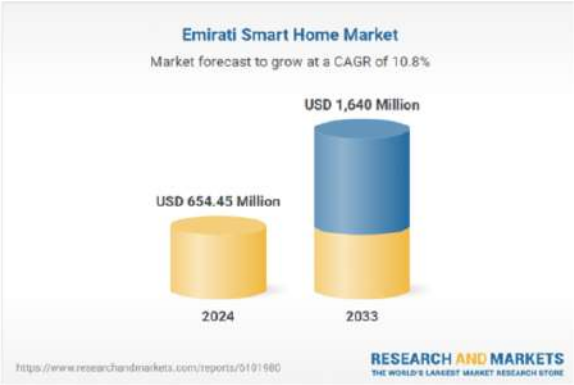


Figure 1: Projected Growth of the Smart Home Market in the United Arab Emirates (2024–2033)

Recent market research further highlights the rapid growth of the smart home industry in the United Arab Emirates, where increasing demand for energy-efficient, secure, and automated living environments is driving significant adoption of smart home technologies. This trend is further illustrated in figure 1, which shows the projected growth of the smart home market in the United Arab Emirates. (Research and Markets, 2024)

2) PROJECT BRIEF AND SCOPE

2.1) Estate description

The project scope involves the design and evaluation of smart residential estate commissioned by an architectural firm. The estate consist of :

- 6x 4-bedroom Family homes
- 8x 2-bedroom affordable starter homes
- Sheltered accommodation with a warden for elderly residents

All properties will be equipped with solar energy systems as a standard feature. the objective is to integrate appropriate smart technologies across the estate to enhance security , energy efficiency , automation and user convenience while considering the specific needs of different occupant groups.

2.2) Assumptions

- The following are the assumptions for this project :
- The project is based in United Arab Emirates
- All costings will be calculated in AED
- The estate has access to high-speed broadband connectivity
- Solar panels are pre-installed in all homes
- Smart devices are compatible with selected platforms (Amazon Alexa, Smart Guard Application)
- The project focuses of design and evaluation not physical implementation

2.3) Deliverables

The project will produce the following deliverables :

- A comprehensive Group report .
- Smart home system designs for each housing type.
- Architecture and integration diagrams.
- Cost analysis table for smart technologies.
- Project management documents , including : PID , WBS, Gantt chart , 4-fields map ,BOQ and Critical path diagram.
- Presentation slides for project demonstration.
- Supporting documentation such as meeting minutes and individual reflections.
- Smart guard Application Interface

2.4) Scope boundaries

Included in Scope:

- Design smart systems for all housing types
- Selection of devices, sensors, and platforms
- Integration of smart technologies and automation
- Energy optimization through solar system
- Identification of integration challenges and solutions

Excluded from scope :

- Physical installation of devices
- Construction of buildings
- Real-world deployment and testing
- Maintenance and operational support

3) STAKEHOLDER ANALYSIS

A stakeholder analysis needs to be performed in order to identify every person or group engaged in or impacted by the smart home project. Comprehending the needs of stakeholders guarantees that the suggested smart home solutions are appropriate, efficient, and in line with user expectations for all types of properties, including starter houses, family homes, and elderly warden.

Stakeholder	Interest in Project	Influence	Requirements
Architects	Designing and planning of estate	High	Smart systems that are practical, efficient, and up to date
Families	Living in the 4 bedroom houses	High	Safety, child monitoring, energy efficiency
Warden Staff	Accommodating the elderly	High	Reliable systems, cost efficiency, safety monitoring
Elderly Residents	Users of assisted living homes	Medium	Reliable systems, cost efficiency, safety monitoring
Maintenance Teams	Install and maintain systems	Medium	Easy installation, system compatibility
Future buyers	Potential occupants	Low-Medium	Affordable, user friendly smart features

Table 1 : Stakeholder table

3) PLATFORMS

The central system that links and manages all of the smart devices in the housing estate is a smart home platform. To guarantee compatibility, usability, security, and successful integration across all three property types such as family houses, senior housing, and starter homes , it is crucial to select the appropriate platform.

3.1) Amazon Alexa :

- Amazon Alexa was picked because of its excellent third-party support and great compatibility with a variety of smart devices.
- Can be suitable for both families and elderly users due to its simple voice control.
- Cost effective and scalable.

3.2) Smart Guard Application Interface :

The Smart Guard Application is the central platform used to monitor and control all smart home systems across the estate. It provides a unified mobile interface that connects with the device, edge, and cloud layers, enabling real-time access, control, and automation of connected devices.

The application offers users to manage security, energy usage, and smart devices from one platform. Role-based access ensures that different users, such as homeowners and wardens, have appropriate levels of control.

- **Security Monitoring and Alerts:**
  Provides real-time CCTV access, motion detection, and intrusion alerts, with AI-based filtering to prioritise critical events.
- **Energy Management:**
  Integrates with solar and monitoring systems to track energy usage and optimise appliance operation during peak generation.

- **Device Control and Automation:**
  Enables remote control of lighting, appliances, and locks, with configurable automation such as scheduled or motion-based actions.
- **Health and Safety Monitoring (Elderly Homes):**
  Supports fall detection, abnormal health alerts, and emergency notifications for rapid response.
- **Garage and Access Control:**
  Allows monitoring and control of smart locks and garage systems, including access permissions and vehicle logs.
- **Notifications and Incident Management:**
  Delivers real-time alerts for system events, faults, and emergencies to ensure immediate user awareness.

Overall, the Smart Guard Application enhances usability, system efficiency, and remote accessibility, making it a critical component of the smart home ecosystem across all housing types.

4) PROJECT METHODOLOGY

This project adopts a structured project management approach to ensure effective planning, coordination, and execution of tasks . A combination of Agile methodology and Scrum framework is used to support iterative development , flexibility, and continuous improvement throughout the project lifecycle .

4.1) PROJECT INITIATION DOCUMENT (PID)

4.1.1) Project Objectives

The key objectives of the project include:

- Design smart home solutions for all three housing types
- Identify suitable smart devices, sensors, and platforms
- Develop a layered system architecture (device, edge, cloud, application)
- Integrate automation workflows and energy management systems
- Evaluate integration challenges and propose solutions
- Ensure security, scalability, and cost-effectiveness

4.1.2) Project Scope

Included in Scope:

- Smart system design for all housing types
- Selection of devices, platforms, and communication protocols
- Integration of automation and IoT systems
- Energy optimisation using solar systems
- Identification of integration challenges and solutions

Excluded from Scope:

- Physical installation of systems
- Construction of buildings
- Real-world deployment and testing
- Maintenance and operational management

4.1.3) Project Deliverables

The project will produce:

- A comprehensive group report
- Smart home architectural and system designs
- Cost analysis (Bills of Quantities)
- Integration and data flow diagrams
- Project management documents (PID, WBS, Gantt Chart, Critical Path,4 fields map , BOQ)
- Presentation slides and supporting documentation

4.1.4) Project Team Structure

The project follows an Agile-Scrum structure:

- Product Owner**:** Defines requirements and ensures alignment
- Scrum Master: Facilitates workflow and removes blockers
- Development Team: Responsible for research, design, and documentation

Collaboration is maintained through:

- Weekly meetings
- Task allocation
- Continuous feedback and review cycles

4.1.5) Stakeholders

Key stakeholders include:

- Architects – design feasibility and sustainability
- Families – comfort, security, and usability
- Elderly residents – safety and independent living
- Warden staff – monitoring and management
- Technology providers – system implementation

Assumptions:

- Project is based in the UAE
- Costs are calculated in AED
- High-speed internet is available
- Smart devices are compatible with selected platforms
- Solar panels are pre-installed

Constraints:

- Limited project timeline
- No access to physical smart devices
- Budget limitations for device selection
- Academic submission requirements

4.1.6) Risk Management

Risk management is applied to identify and address potential risks that may affect the successful delivery of the smart home project. These include technical, financial, and operational risks related to system integration, device compatibility, and security.Risks are assessed based on their likelihood and impact, allowing prioritisation of critical issues. Mitigation strategies are defined to minimise disruptions and ensure project continuity.A risk register is maintained to document and monitor risks throughout the project lifecycle.

Risk	Mitigation
Device Compatibility Issues	Use of standard protocols (eg:Zigbee, WiFi)
Device Compatibility issues	Data filtering and prioritization
Network delays	Edge computing
System complexity	Centralized smart home platforms

Table 2 : Risk management table

4.1.7) Success Criteria

The project will be considered successful if:

- All housing types have appropriate smart designs
- Energy optimisation through solar integration is clearly demonstrated
- Costs are realistic and justified
- Security and safety features are effectively implemented
- The report meets professional and academic standards

4.2) Project Approach

The project follows an agile methodology supported by Scrum practices, allowing the team to work in iterative cycles.Tasks are divided into manageable units and assigned to team members based on responsibilities.
Regular team meetings,progress reviews, and feedback sessions ensure:

- Continuous improvement
- Efficient task management
- Adaptability to changes

This approach enhances collaboration and ensures timely completion of project deliverables.

4.3) Diagrams

The Work Breakdown Structure (WBS) breaks the project into smaller, manageable tasks to support planning, task allocation, and progress tracking. Each activity is assigned a hierarchical code to show its position within the overall project. This coding system improves task organisation and supports integration with the Gantt chart and critical path.

4.3.1) WORK BREAKDOWN STRUCTURE (WBS)

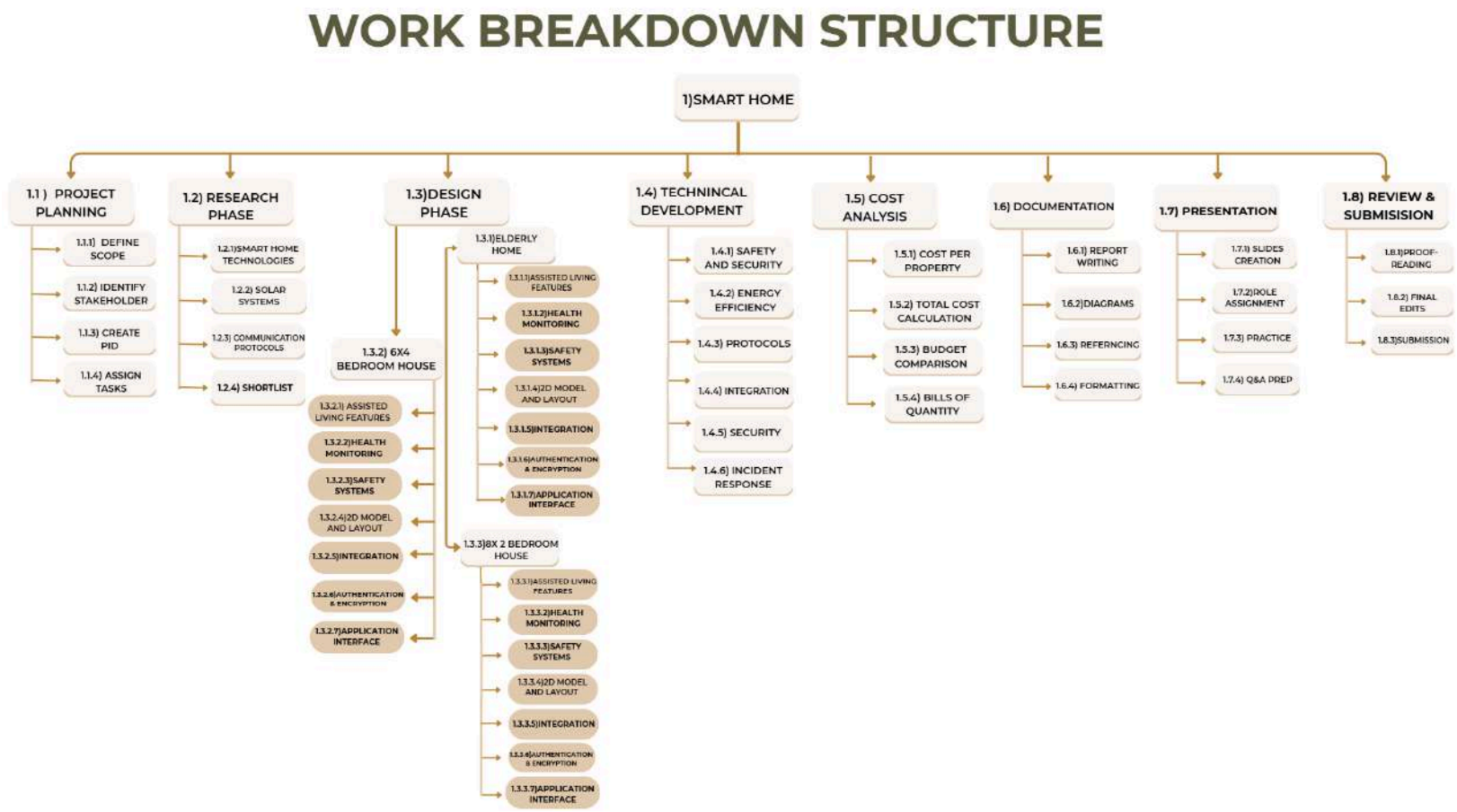


Figure 2 : Work breakdown structure

4.3.2) Gantt Chart

Provides a timeline of activities, ensuring proper scheduling and task sequencing

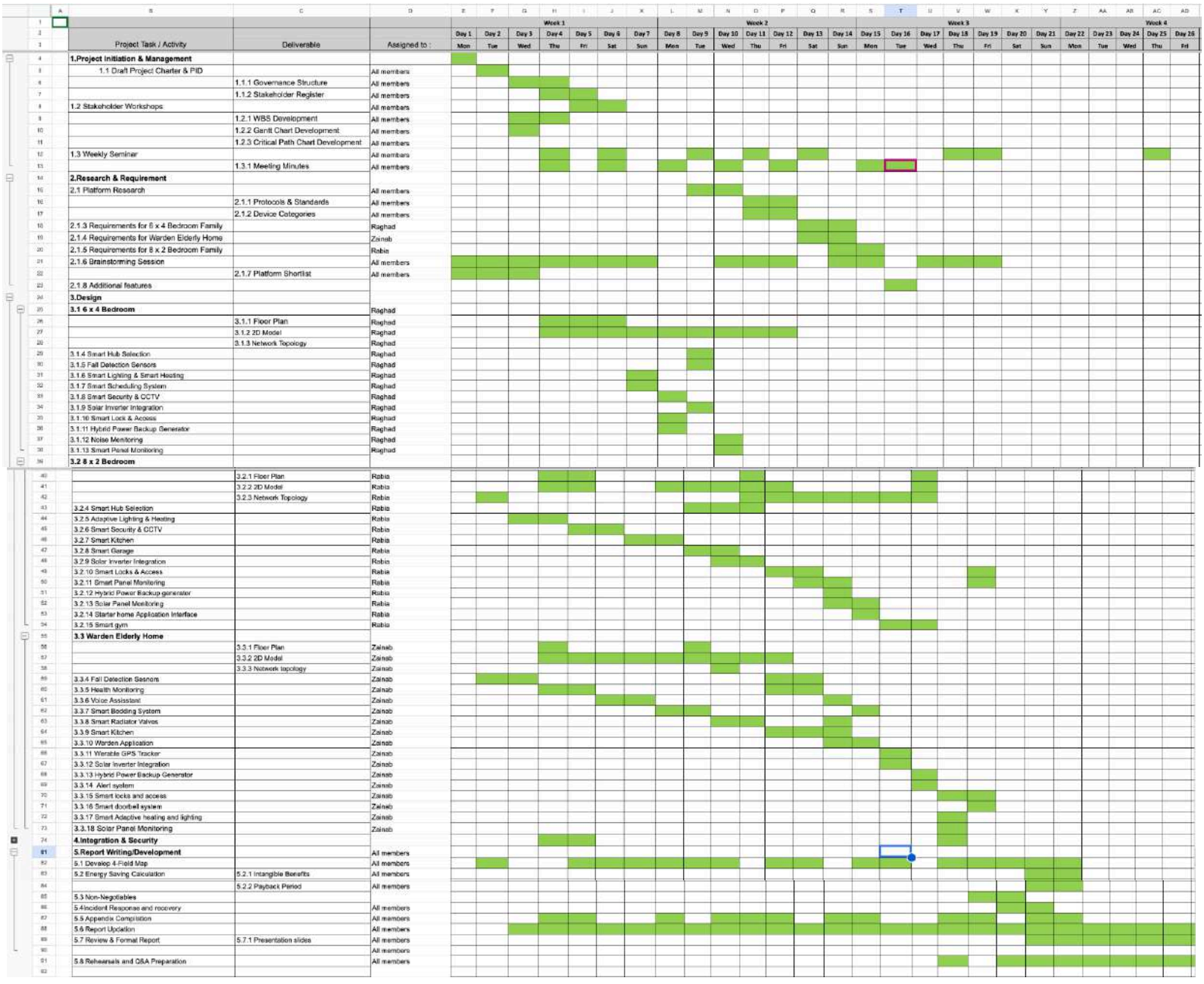
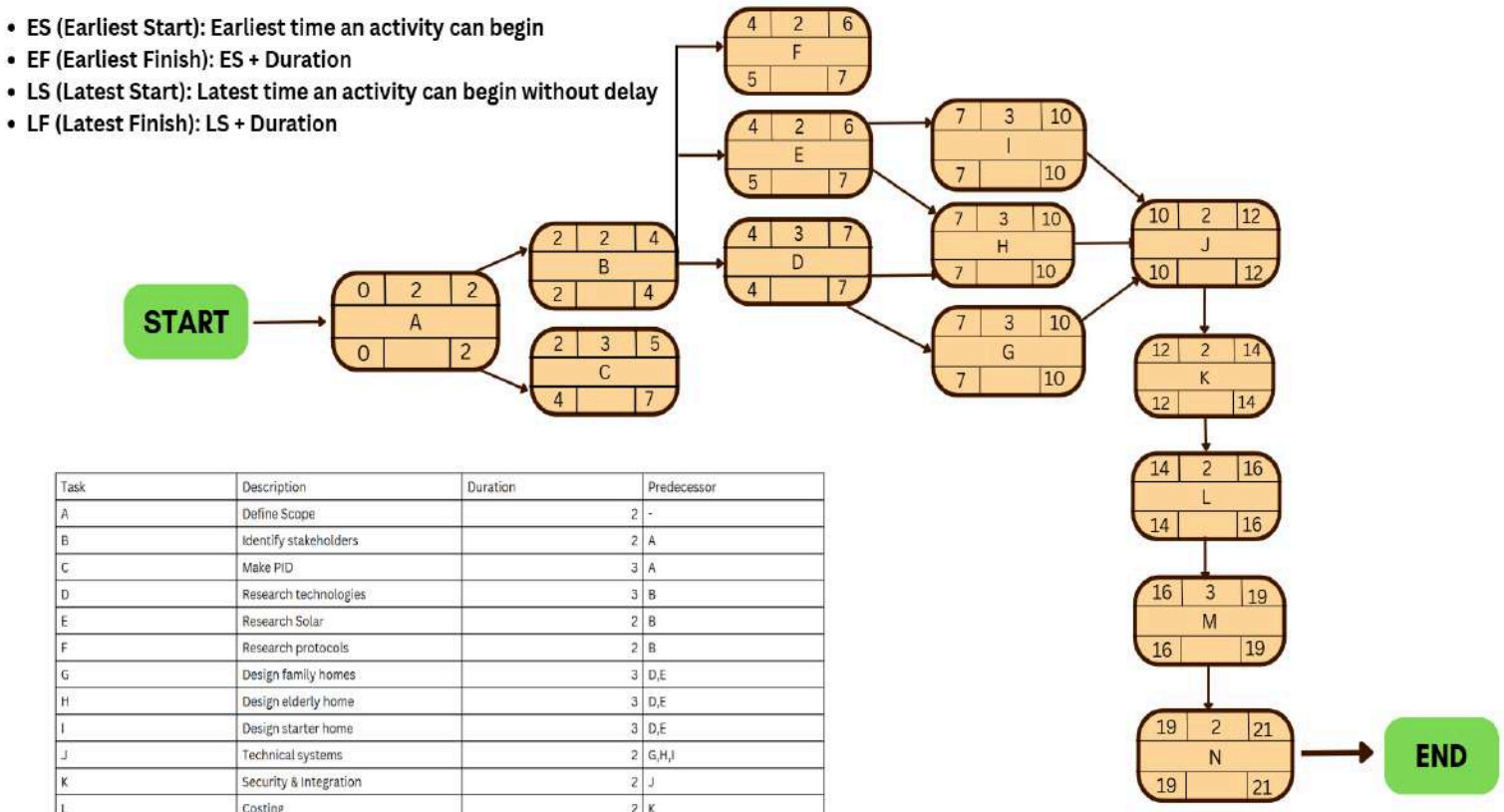


Figure 3 : Gantt Chart

4.3.3) Critical path diagram

Identifies the sequence of dependent tasks that determine the minimum completion time of the project, ensuring that key activities are prioritised and delays are avoided.



Task	Description	Duration	Predecessor
A	Define Scope	2	-
B	Identify stakeholders	2	A
C	Make PID	3	A
D	Research technologies	3	B
E	Research Solar	2	B
F	Research protocols	2	B
G	Design family homes	3	D,E
H	Design elderly home	3	D,E
I	Design starter home	3	D,E
J	Technical systems	2	G,H,I
K	Security & Integration	2	J
L	Costing	2	K
M	Report writing	3	L
N	Presentation	2	M

Figure 4: Critical Path Diagram

4.3.4) Stakeholder Power Matrix

	← High Interest	Low Interest →
High Power ↑	**Q1 - Manage Closely** *High Power | High Interest* Key decision-makers and champions. Engage frequently, involve in planning, and keep fully informed. Their support is critical to project success. • **Project Sponsor** - Final approver of budget & scope • **Homeowner Client** - Primary end-user and decision-maker • **Lead Architect** - Defines system design standards • **University Supervisor** - Academic oversight & marking • **IoT Platform Vendor** - Core tech dependency • **Security Compliance Officer** - Approves safety protocols • **Smart Home Integration Lead** - Technical authority • **Budget Controller** - Approves major expenditures	**Q2 - Keep Satisfied** *High Power | Low Interest* Influential stakeholders who are not deeply involved day-to-day. Keep informed at a high level, manage expectations, and avoid surprises. • **University Ethics Board** - Approves data privacy • **Local Government / Council** - Permits & compliance • **Insurance Provider** - Liability & coverage decisions • **Senior Management** - Strategic alignment • **IT Infrastructure Manager** - Network & server access • **Legal / Compliance Team** - Contractual sign-off • **Property Developer** - Site access & agreements • **Funding Body / Grant Committee** - Release of funds
Low Power ↓	**Q3 - Keep Informed** *Low Power | High Interest* Engaged stakeholders with limited authority. Keep them updated with regular communications to maintain goodwill and gather useful feedback. • **End Users / Residents** - Daily system interaction • **Elderly Care Workers** - Monitor assisted living features • **Maintenance Technicians** - Ongoing system upkeep • **Student Team Members** - Task execution & delivery • **Community Representatives** - Neighbourhood feedback • **Health & Safety Advisors** - Review safety measures • **Warden / Building Manager** - On-site operations • **Sustainability Advocates** - Energy efficiency goals	**Q4 - Monitor** *Low Power | Low Interest* Peripheral stakeholders. Monitor periodically and provide general updates. Do not invest significant engagement resources. • **General Public** - Broad awareness only • **Media / Press** - Occasional coverage • **Competing Smart Home Vendors** - Market context • **Academic Peer Reviewers** - Post-project evaluation • **Adjacent Property Owners** - Passive neighbours • **Utility Providers** - Baseline infrastructure • **Product Distributors** - Supply chain periphery • **Future Prospective Clients** - Long-term pipeline

Figure 5 : Stakeholder power matrix

Legend & Engagement Strategy

Q1 - Manage Closely	High Power, High Interest. Decision-makers and champions - engage deeply, involve in all key decisions.
Q2 - Keep Satisfied	High Power, Low Interest. Influential but not hands-on - brief regularly, manage expectations, avoid surprises.
Q3 - Keep Informed	Low Power, High Interest. Engaged supporters - share updates, gather feedback, maintain positive relationships.
Q4 - Monitor	Low Power, Low Interest. Peripheral stakeholders - periodic updates only, no significant engagement needed.

Table 3 : Legend engagement strategy

4.3.5) 4-Fields Map

Identifies strengths, weaknesses, opportunities, and threats to support strategic decision-making.

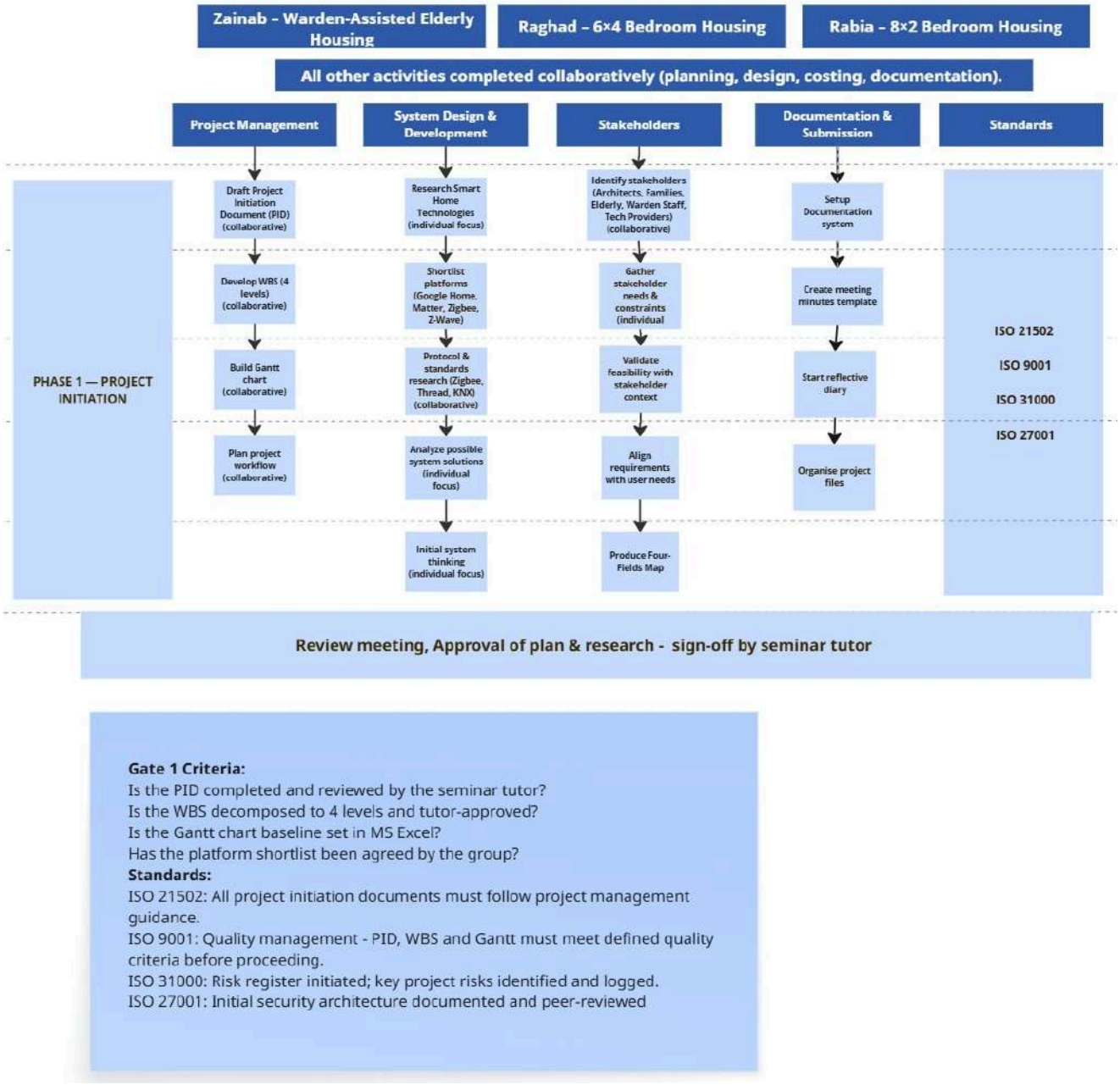


Figure 6 : Phase 1 , 4-Fields map

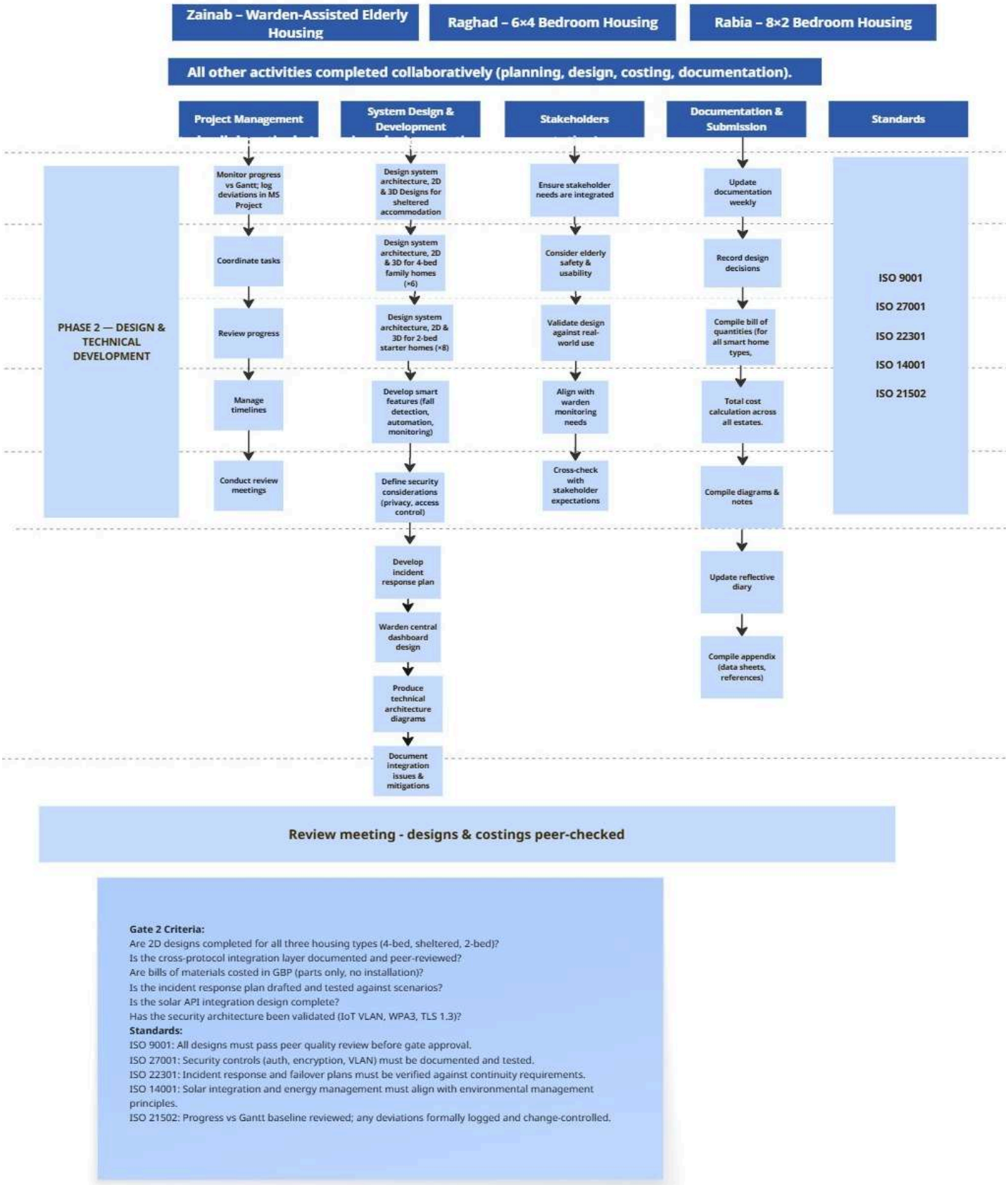
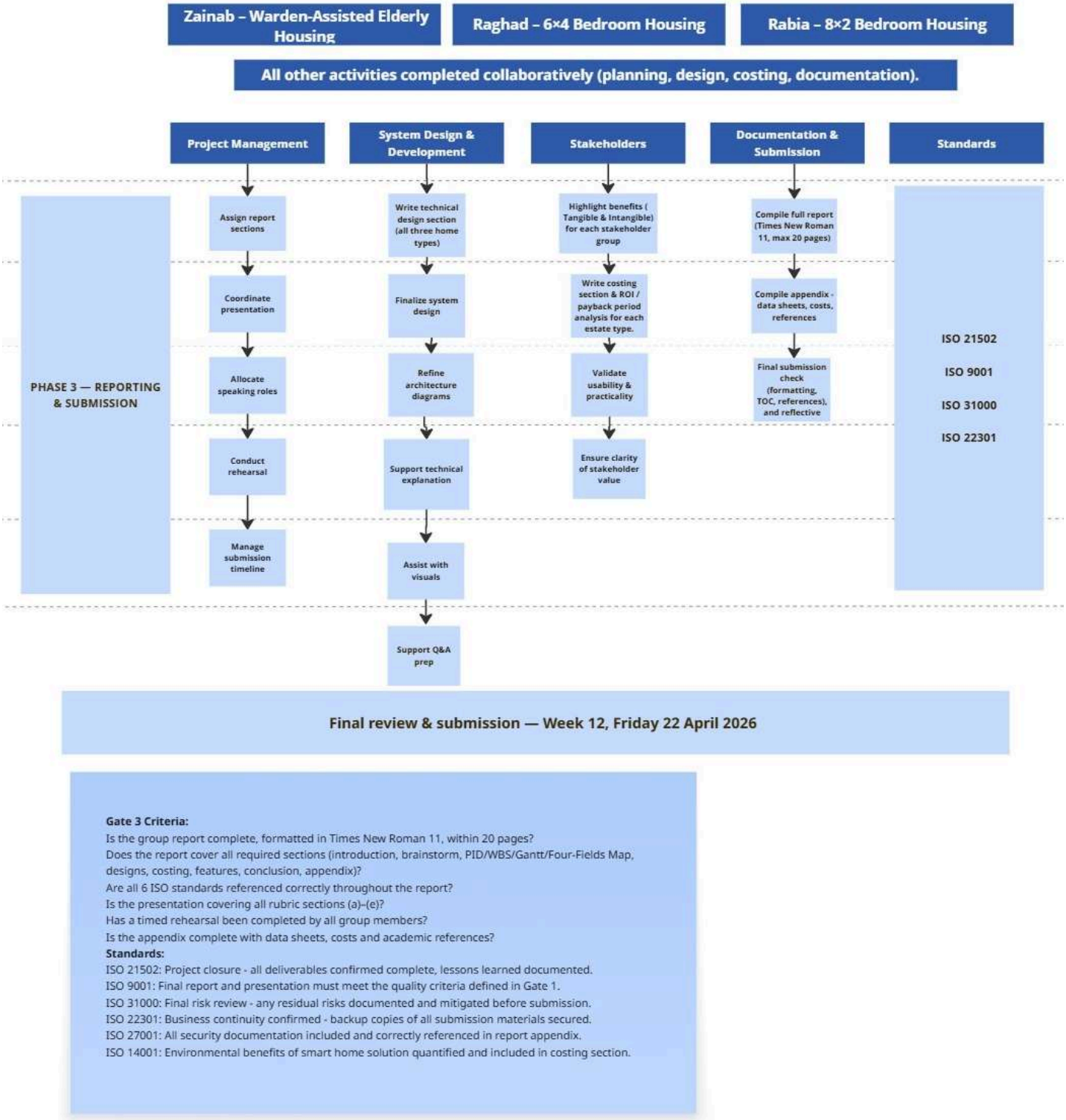


Figure 7 : Phase 2 , 4-Fields map



Gate 3 Criteria:
Is the group report complete, formatted in Times New Roman 11, within 20 pages?
Does the report cover all required sections (introduction, brainstorm, PID/WBS/Gantt/Four-Fields Map, designs, costing, features, conclusion, appendix)?
Are all 6 ISO standards referenced correctly throughout the report?
Is the presentation covering all rubric sections (a)–(e)?
Has a timed rehearsal been completed by all group members?
Is the appendix complete with data sheets, costs and academic references?
Standards:
ISO 21502: Project closure - all deliverables confirmed complete, lessons learned documented.
ISO 9001: Final report and presentation must meet the quality criteria defined in Gate 1.
ISO 31000: Final risk review - any residual risks documented and mitigated before submission.
ISO 22301: Business continuity confirmed - backup copies of all submission materials secured.
ISO 27001: All security documentation included and correctly referenced in report appendix.
ISO 14001: Environmental benefits of smart home solution quantified and included in costing section.

Figure 8 : Phase 3 , 4-Fields Map

4.4) Decision-making process

The project follows a collaborative decision-making approach, where key decisions are made through group discussions and consensus.
The decision-making process includes:

- Identification of requirements and constraints
- Evaluation of alternative solutions
- Comparison based on feasibility, cost, and performance
- Selection of the most suitable option

For technical decisions (e.g., platform selection or device integration), priority is given to:

- Compatibility
- Cost-effectiveness
- Scalability
- Security

This structured approach ensures that all decisions are justified, practical, and aligned with project objectives.

4.5) Risk management approach

Risk management is implemented to identify , assess, and mitigate potential issues that may impact the project ,risks are continuously monitored throughout the project, and mitigation strategies are applied to minimize their impact.

Risk	Impact	Mitigation strategy
Device Compatibility issues	HIGH	Use a unified IoT platform (e.g., Google Home/Alexa) and select devices that support standard protocols like Zigbee and WiFi
Data overload from sensors	HIGH	Use AI-based filtering and prioritisation to ensure only important alerts are delivered
Network delays	MEDIUM	Implement edge computing to process critical data locally and reduce reliance on cloud communication
System complexity	MEDIUM	Use a centralised mobile application to control all devices through a single interface
Security Vulnerabilities	MEDIUM	Apply encryption, authentication, and access control mechanisms to secure communication and data

Table 4 : Risk management

5) SHARED TECHNICAL FRAMEWORK FOR THE ESTATES

This section outlines the common technical architecture used across all housing types, enabling seamless integration of devices, data flow, and automation.

5.1) Protocols

For smart devices to be able to interact with one another, communication protocols are crucial. These protocols guarantee effective, safe, and dependable communication between gadgets like cameras, sensors, and smart appliances.

- **Wi-Fi:**
  The main communication protocol used to connect high-bandwidth devices like voice assistants, smart cameras, and mobile apps is Wi-Fi. Homeowners may monitor and manage their systems from any location thanks to its real-time data transmission and remote access capabilities.
- **Bluetooth:**
  Short-range communication between devices is done by Bluetooth, especially during initial setup and device pairing. It removes the need for a continuous internet connection while enabling fast and secure connectivity.
- **Zigbee and Z-wave:**
  Low-power wireless communication protocols like Z-Wave and Zigbee are used by home automation devices, lighting systems, and smart sensors. Both protocols are appropriate for equipment that need to run continuously and are energy-efficient.While Zigbee allows for more connected devices within the network, Z-Wave provides dependable connectivity with less interference.

5.2) Security

- ***User Authentication and Control:***
  Secure authentication techniques like multi-factor authentication (MFA) and strong passwords safeguard access to the smart home system. This guarantees that devices and system data can only be accessed by authorized users. It is also possible to use role-based access, which gives parents complete control while limiting children's access.
- ***Secure Wi-Fi Network:***
  To reduce the possibility of cyberattacks and enhance network security, strong encryption technologies such as WPA2,WPA3 are used to keep the smart devices isolated from personal devices by creating a distinct network.
- ***Data Encryption:***
  To prevent any unauthorized interception of sensitive information, all data that is being transmitted between smart devices, mobile applications and cloud devices are encrypted using secure protocols.
- ***Device authentication and updates:***
  Before connecting any of the smart devices to the network, authentication needs to be done. Regular software updates are also applied to ensure that devices are protected against any vulnerabilities.

5.3) Sensors

Smart sensors form the foundation of the smart home system by collecting real-time environmental, security, and user-related data. A shared set of sensors is implemented across all housing types, with variations in quantity and usage depending on specific requirements.

Sensor Type	Use Case	Application
Motion Sensors	Detects movement	Security Alerts , Lighting automation
Occupancy sensor	Detects occupancy	Energy optimization
Temperature Sensor	Monitors temperature	Smart thermostat control
Smoke / CO sensor	Detects fire and gas leaks	Safety and emergency alerts
Door / Window Sensor	Detects opening and closing	Intrusion detection , automation and remote control
Energy Monitor	Tracks electricity usage	Solar energy management
Garage sensors	Detects the vehicles entry and exit	Vehicle logging system
Wearable Health sensor	Vital monitor	Health monitoring

Table 5 : Sensors and its application

5.4) Standards

ISO 21502: Project, programme and portfolio management

To ensure rigorous management of our smart home project, we adopted ISO 21502:2020 Project Management Guidelines, which provide a structured approach to governance, lifecycle stages and continuous improvement (ISO, 2020). This framework guided the planning, execution, and mentoring of activities including integration of IoT devices, automated lighting, security systems, and renewable energy solutions. By celery defining roles and responsibilities, we ensured accountability and informed decision-making across all stakeholders. Risk management measures were implemented to mitigate cybersecurity threats and system failures, fully aligned with UAE data protection laws and smart technology regulations. Moreover, our approach supports UAE Vision 2031 and Smart Dubai initiatives by promoting sustainability, energy efficiency, and innovation. Continuous evaluation and lessons learned have further enhanced project efficiency and compliance with both international and national standards (ISO, 2020).

ISO 27001 - Information Security Management Systems

In our Smart Home project, ISO/IEC 27001:2023+A1:2024 serves as the core information security standard, providing a structured framework to safeguard all aspects of the system. The standards' organizational, people, physical, and technological controls ensure the confidentiality, integrity, and availability of sensitive information processed by IoT devices, smart hubs, cloud services, and user interfaces. Implementation of ISO27001 principles enables secure device configuration, encrypted communications, controlled access management, regular monitoring, and incident response planning. Additionally, it supports compliance with UAE data privacy regulations and enhances the resilience of smart home operations against cyber, physical, and environmental threats. By embedding ISO 27001 controls into the project lifecycle, the smart home systems achieves high operational security, reliability, and user trust.

ISO 31000 Risk Management

By taking into account both internal and external elements that could impact system performance across all 3 properties, the ISO 31000 principles are used for the smart home project. Internal elements like project objectives, resources, and system capabilities, external factors like legal, technological, and environmental conditions are considered. Strong stakeholder commitment, where duties are clearly defined and resources are distributed to successfully manage risks, is essential to risk management. In order to ensure that risks like system failures, cybersecurity threats, and operational interruptions are constantly monitored and mitigated, the strategy is fully incorporated into the planning, design, and implementation stages of starter houses, family homes, and senior housing. This guarantees a smart home system that is organized, dependable, and consistent throughout the whole housing estate. (ISO 31000, 2018)

ISO 14001 Environmental Management

The smart housing project promotes environmental sustainability in all three categories of properties by implementing ISO 14001 principles.

- To minimize waste and maximize resource utilization, the standard is implemented by incorporating energy-efficient technology including solar panels, smart energy monitoring systems, and automated energy control.
- To guarantee lower energy use and compliance to environmental laws, environmental performance is constantly assessed.
- This strategy lowers operating costs and increases overall efficiency while promoting sustainable design in starter houses, senior housing, and family homes. Through ethical and sustainable smart home solutions, the initiative improves environmental performance, facilitates regulatory compliance, and fosters stakeholder trust. (ISO, 2026)
-

ISO 9001 Quality Management

- ISO BS EN ISO 9001:2015+A1:2024 has been applied to our smart home project to ensure a structured, quality-focused approach across the design, procurement, installation, and maintenance of smart technologies (IBS EN ISO 9001:2015+A1:2024). As our project relies on third-party vendors for IoT devices, CCTV systems, and automation platforms, strict control over externally provided products and services is maintained through clearly defined technical and compliance requirements, alongside continuous supplier evaluation (BS EN ISO 9001:2015+A1:2024).
- We implement standardised installation procedure, utilise skilled personnel, and conduct systematic testing to ensure controlled service provision, while identification and traceability mechanisms enable effective maintenance and fault resolution (BS EN ISO 9001:2015+A1:2024). Furthermore, we integrate risk-based thinking to identify and correct nonconforming outputs at an early stage, ensuring only reliable and compliant systems are deployed (IBS EN ISO 9001:2015+A1:2024).
- We also carry performance evaluation through user feedback, system performance metrics, internal audits, and management reviews to ensure alignment with our project objectives, customer expectations, and UAE regulatory frameworks and smart city initiatives (UAE Government, 2021). Ultimately, ISO 9001 supports our commitment to continuous improvement, enabling our smart home system to evolve with emerging technologies while ensuring long term efficiency, reliability, and customer satisfaction.

ISO 22301 Business Management

- In our smart home project, ISO 22301:2019+A1:2024 is adopted to ensure continuity, resilience, and reliability of critical operations, particularly during system failures, cyber incidents, or infrastructure disruptions. This is especially important given the focus on elderly users, where service interruptions may directly impact safety and well-being (BS EN ISO 22301:2019+A1:2024).
- The standard is implemented through a structured, risk-based approach involving Business Impact Analysis (BIA) and risk assessment to identify and prioritise critical components such as smart security systems, IoT sensors, automated bedding, and energy management systems. Key metrics, including Maximum Tolerable Period of Disruption (MTPD) and Recovery Time Objectives (RTOs), are defined to ensure near-zero downtime for critical services like emergency alerts, while allowing longer recovery periods for non-critical functions.
- To enhance resilience, redundancy mechanisms such as cloud backups, alternative connectivity, and manual overrides are integrated alongside clearly defined incident response and recovery procedures. This approach aligns with UAE Vision 2031 and the UAE Digital Government Strategy 2025, which emphasise digital resilience and smart infrastructure development.
- Additionally, regular testing, monitoring, and continuous improvement processes are incorporated to evaluate system performance and address emerging risks, ensuring long-term adaptability (BS EN ISO 22301:2019+A1:2024). While implementation may increase system complexity and cost, ISO 22301 significantly strengthens system robustness, safety, and sustainability. When combined with standards such as ISO/IEC 27001 and ISO 9001, it contributes to a comprehensive framework for secure, resilient, and high-quality smart home environments.

Legal Considerations

The proposed smart home estate operates within the UAE's legal and regulatory framework, requiring compliance with data protection, cybersecurity, surveillance, and safety laws. The system adheres to the UAE Federal Decree-Law No. 45 of 2021 on Personal Data Protection, ensuring lawful processing of personal and health-related data through user consent, data minimisation, and secure storage.

- It also complies with UAE Federal Decree-Law No. 34 of 2021 on Cybercrimes and guidance from the UAE Cybersecurity Council, enforcing strong cybersecurity controls such as encryption, secure networks, and incident response, aligned with ISO 27001. Surveillance systems (e.g., AI-enabled CCTV) follow regulations by SIRA and Dubai Police to protect privacy and restrict data access.
- Elderly care and health monitoring systems align with Ministry of Health and Prevention (UAE) requirements, supporting data protection and system reliability, in line with ISO 22301. Additionally, smart devices comply with consumer protection laws under the Ministry of Economy (UAE), reinforcing ISO 9001 principles.
- All communication technologies meet Telecommunications and Digital Government Regulatory Authority standards for secure IoT connectivity. The integration of solar energy and automation aligns with UAE Vision 2031 and the Dubai Clean Energy Strategy 2050, supporting sustainability goals under ISO 14001.

Overall, integrating these legal and regulatory requirements ensures compliance, reduces risks, and enhances user trust, supporting the feasibility of real-world deployment in the UAE.

5.5) Incident Response & Recovery

In the event of a fault, damage, or emergency situation, the smart home system incorporates a structured incident response and recovery plan. This plan is designed to minimise disruption, protect occupants, and restore normal system functionality through the use of automated alerts, backup power systems, and predefined recovery procedures.

Incident Response	***Recovery***
Power Failure	Important devices like security cameras, sensors, and communication systems will continue to function in the event of a power outage because of the hybrid backup system, which consists of a generator and solar battery. By doing so, safety monitoring is maintained and system downtime is avoided.
Emergency Alerts	In case of incidents such as gas leaks, fires, falls or unauthorized access the system will automatically detect this and send real time information to the homeowners.
Data Backup	Device configurations and user settings, among other crucial system data, are safely kept and backed up. Data can be promptly restored to minimize disruption in the case of a system failure or reset
Security Incidents	The system can alert the user and try to automatically reconnect if a smart device or system malfunctions. Along with that, there are manual override options that let people control necessary appliances (such heating or lighting) without depending on automation.
Maintenance Updates	To avoid any malfunctions and guarantee that every equipment functions effectively, routine system updates and maintenance inspections are performed. This lessens the possibility that mishaps will happen.
System Failure and recovery	The system can alert the user and try to automatically reconnect if a smart device or system malfunctions. Along with that, there are manual override options that let people control necessary appliances (such as heating or lighting) without depending on automation.

Table 6 : Incident response and recovery

5.5.1) Scenario

Power failure

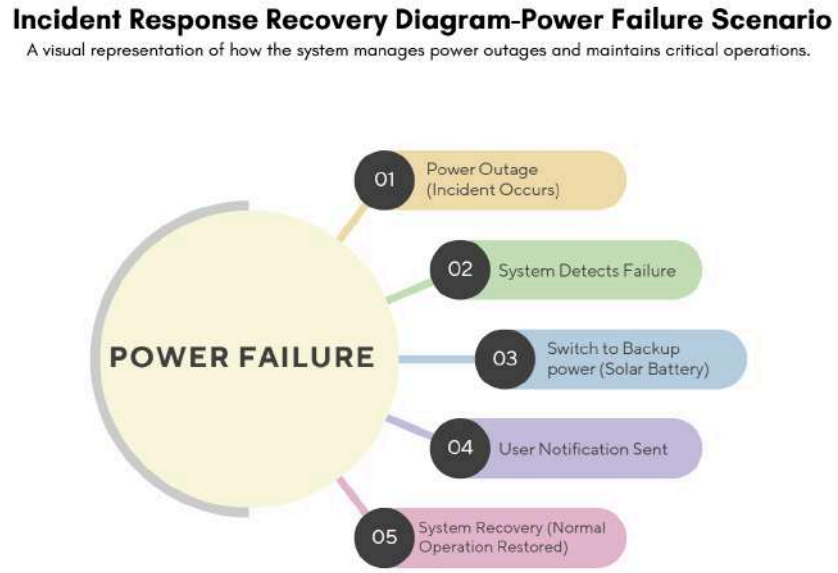


Figure 9 : Incident Response & Recovery Diagram

Cyber Attack

In the context of the United Arab Emirates, where smart home technologies are rapidly being adopted as part of smart city initiatives and digital transformation strategies, ensuring robust cybersecurity measures is essential. Smart home systems rely on interconnected IoT devices, cloud services, and real-time data exchange, which increases their exposure to cyber threats such as unauthorized access, data breaches, and system disruptions. Therefore, implementing a structured incident response and recovery process is critical to maintaining system reliability, protecting user data, and ensuring continuous operation. The following flowchart illustrates a standardised approach to managing cyberattacks in smart home environments, this incident response and recovery process is based on established cybersecurity frameworks such as NIST Figure : guidelines and ISO 27001 standards, which define structured approaches to detecting, responding to, and recovering from cyber incidents (NIST, 2012; ISO, 2022).

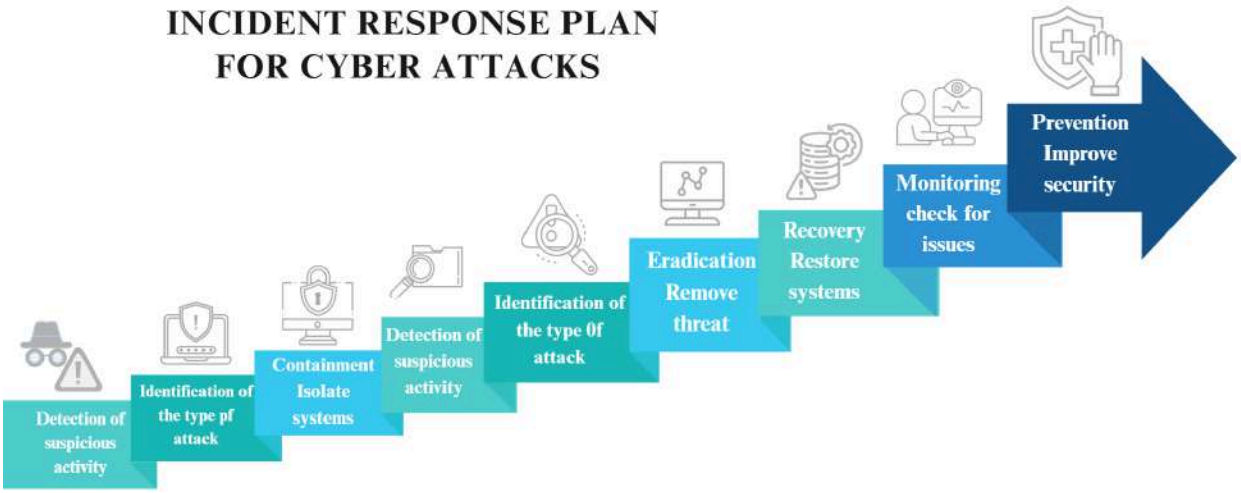


Figure 10 : Incident Response Plan for Cyber Attacks

5.6) Integration

To guarantee that every system in the works has a whole intelligent ecosystem, smart device integration is very crucial. A fundamental smart home platform connects several smart technologies in the planned architecture allowing smooth communication, automation and economical energy of use.
The following integrations are implemented:

- **Solar Energy Integration:**
  The solar monitoring system is integrated with smart home devices to optimize energy usage. When high levels of solar energy are available, the system can automatically activate appliances or increase usage efficiency. For example, energy-intensive devices can operate during peak solar generation periods, reducing reliance on external electricity sources.
- **Mobile Application Integration:**
  Through a single mobile application, users can access all smart devices, remotely control and monitor devices. The app ensures that the homeowners are always connected to their property by providing real time information, notifications and system status.
- **Central Smart Home Platform Integration (Alexa):**
  A voice assistant system like Amazon Alexa connects all smart gadgets in the house via a central platform. This allows voice commands or a mobile application to manage lighting, sensors, cameras, and other devices from a single interface. This connection improves user simplicity and improves managing the system.
- **Network and Protocol Integration:**
  Device connectivity is supported by communication protocols like Bluetooth, Wi-Fi, Zigbee, and Z-Wave. While low-power devices like sensors use Zigbee or Z-Wave, high-bandwidth devices use Wi-Fi. This combination guarantees scalability, dependable communication, and effective operation throughout the smart home system.

Integration Architecture:

The system integrates devices such as ***Hikvision CCTV cameras, Amazon Alexa voice assistants, smart sensors, Emporia energy monitors, and EcoFlow backup systems*** within a layered architecture:

The architecture consists of four key layers:

- **Device Layer :**
  Includes all physical components such as sensors, smart locks, AI-enabled CCTV cameras with behavioral AI , smart appliances, and wearable devices. These devices collect and generate real-time data.
- **Edge Layer (Gateway/Hub):**
  A central smart hub Alexa processes data locally and enables communication between devices, reducing latency and improving response time.
- **Cloud Layer:**
  Responsible for data storage, analytics, and remote access. It supports advanced features such as AI-based behaviour analysis and system monitoring.
- **Application Layer:**
  A mobile application interface allows users to monitor and control all devices, including security systems, appliances, lighting, and energy usage.

This layered architecture ensures seamless integration of multiple smart devices while maintaining system flexibility and scalability

Integration Challenges and Solutions:

Challenges	***Description***	***Solution***
Device compatibility	Different devices may use different protocols, causing integration issues	Use standard protocols (Wifi, Zigbee, Bluetooth) and a unified IoT platform
Network Reliability	Connectivity issues may affect system performance	Implement stable Wi-Fi networks and backup connectivity solutions
Data Overload	Large Volumes of sensor and CCTV data may overwhelm systems	Use AI-based filtering and prioritisation
Data Security	Risk of unauthorised access and data breaches	Apply encryption, secure authentication, and regular updates
System Scalability	Difficulty in expanding the system over time	Use modular design and cloud-based architecture
Interoperability	Devices from different vendors may not communicate effectively	Use platforms with strong third-party integration support
Power Dependency	System failure during power outages	Use backup systems such as batteries and generators

Table 7 : Integration challenges and solutions table

5.7) Data Flow and Automation

The system operates through continuous data exchange between devices, enabling intelligent automation across the estate.

Data Flow Process

1. Sensors and devices collect real-time data (e.g., motion, temperature, activity)
2. Data is transmitted to the edge layer for initial processing
3. Relevant data is sent to the cloud for analysis and storage
4. Processed data is used to trigger automated actions
5. Notifications and controls are delivered to users via the mobile application

Automation Examples

- Solar Integration:
  Energy monitoring systems detect peak solar generation and automatically activate appliances to optimize energy usage.
- AI CCTV Monitoring (Hikvision):
  Behavioural AI differentiates between normal and suspicious activity, triggering alerts only when necessary.
- Motion-Based Automation:
  Lights and security systems activate automatically based on movement detection.
- Garage System Automation:
  Vehicle entry and exit are logged, and the garage door operates automatically via the mobile application.
- Smart Gym
  Tracks user activity , health monitoring , posture control , provides feedback and statistics via app

Automated data flow reduces manual intervention, improves efficiency, and enhances user experience.

5.8) Energy Efficiency Systems

Energy efficiency is a core component of the smart home design, with systems integrated to optimise energy consumption across the estate.

- Solar Panel Integration:
  All homes are equipped with solar panels to generate renewable energy.
- Smart Energy Monitoring :
  Emporia energy systems Real-time monitoring allows users to track energy usage and identify optimisation opportunities.
- Automated Energy Management:
  Appliances are automatically activated during peak solar energy generation to maximise efficiency.
- Smart Thermostats and Lighting:
  Temperature and lighting systems adjust automatically based on occupancy and environmental conditions.

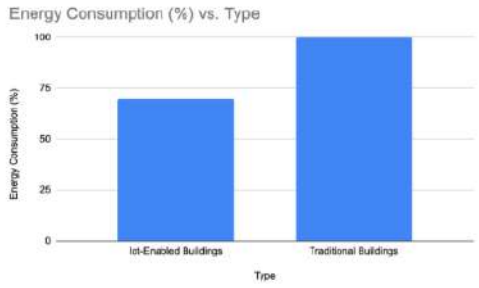


Figure 11: Comparison of Energy Consumption Between IoT- Enabled and Traditional Buildings (%) (World Journal of Advanced Research and Reviews, 2025)

- Hybrid Backup System:
  EcoFlow generators Battery storage and backup systems ensure continuous power supply during outages. These systems reduce energy consumption, lower costs, and support sustainable living within the estate.

6) ESTATES LAYOUT AND DESIGN

This section presents the design and evaluation of smart home solutions for the different housing types within the *proposed* residential estate. Each housing type is designed based on the specific needs of its occupants, ensuring a balance between functionality, cost-effectiveness, energy efficiency, and security.The integration of smart technologies, including IoT devices, automation systems, and energy management solutions, enables each property to operate efficiently while enhancing user comfort and safety. To provide a clear overview, Table 7 presents a comparative analysis of key smart home features across the housing types, highlighting differences in areas such as security, health monitoring, energy efficiency, and overall user requirements.

Category	Elderly Warden Home	6× 4-Bedroom Smart Homes	8× 2-Bedroom Starter Homes
Safety & Security	Fall detection (wearable + room), AI cameras, panic buttons, smart locks (NFC/fingerprint), motion sensors, GPS tracking (dementia) → Faster response, reduced injuries	Enhanced home security systems → Improved resident safety	Enhanced home security systems → Improved resident safety
Health Monitoring	Smart medication dispensers, vital monitoring (HR, BP, sleep) linked to dashboard → Early detection, proactive care	Health-supportive environment (smart living features) → Improved well-being	Health-supportive environment (basic smart features) → Improved well-being
Independence & Daily Living	Voice assistants (e.g., Amazon Alexa), smart beds, automated curtains, kitchen shut-off → Increased independence, reduced caregiver reliance	Improved convenience through automation → Higher comfort levels	Improved convenience through automation → Better accessibility for first-time homeowners
Cognitive & Social Engagement	Smart library (bone-conduction audio, magnification tables, circadian lighting), adaptive workshop tools → Mental stimulation, social interaction	Not a primary focus	Not a primary focus
Energy Efficiency & Cost Savings	Solar panels, solar windows, AI energy optimisation → Reduced energy costs, efficient usage	Reduced electricity costs, efficient energy consumption	Reduced electricity costs, efficient energy consumption
Environmental Sustainability	Renewable energy integration → Lower carbon footprint	Sustainable energy usage → Reduced environmental impact	Sustainable energy usage → Reduced environmental impact
Environmental Health	VOC sensors + automated ventilation → Improved air quality	General improved living environment	General improved living environment
Economic Value	Long-term healthcare cost reduction + operational savings	Increased property value	Increased property value

Table 8 : Comparative Feature analysis

6.1) Elderly home *with* warden

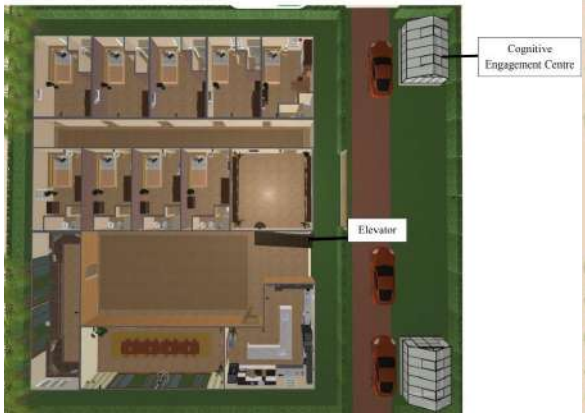
6.1.1) Overview:

The Warden-Assisted Home is designed as an Ambient Assisted Living (AAL) environment, where technology is integrated into daily life in a natural and unobtrusive way, supporting residents without being noticeable. The main aim is to promote independence and improve quality of life for residents, while also providing the warden with a centralized “Single Pane of Glass” dashboard for clear and efficient monitoring of the entire facility. The system combines multiple layers of sensors and smart technologies to address common challenges linked to ageing, particularly falls and cognitive or social isolation. At the same time, the design considers long-term efficiency and sustainability, ensuring that care services remain reliable, practical, and adaptable over time (Khalil et al., 2025).

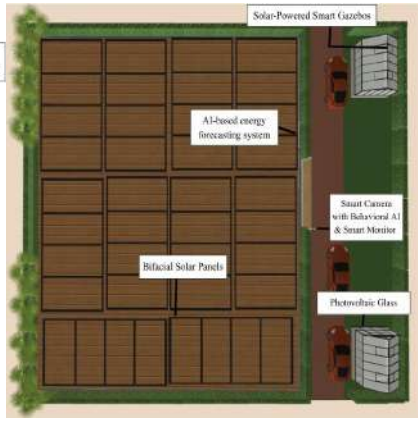
6.1.2) Layout & Design:

Floor 1 & 2 Interior Design (2D):

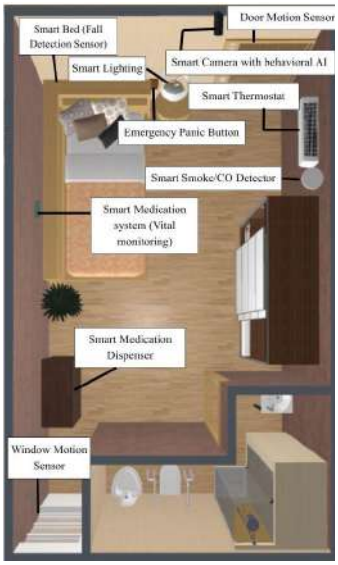
Floor 1 & 2 (3D) Design:



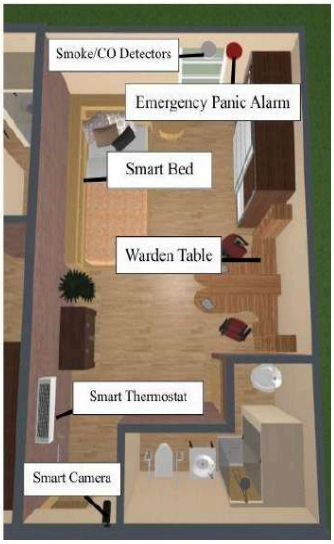
Smart Home Exterior Design:



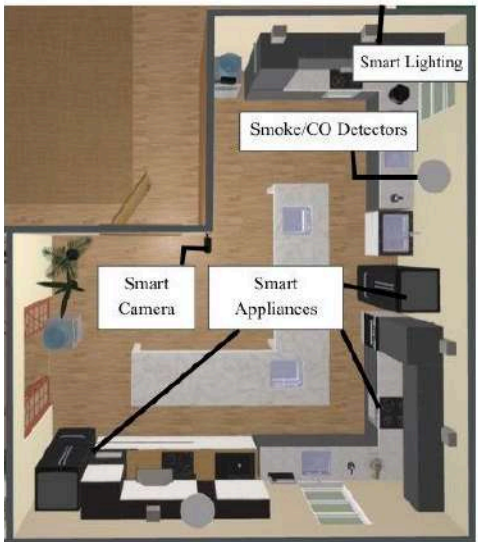
Smart Elderly Room Interior:



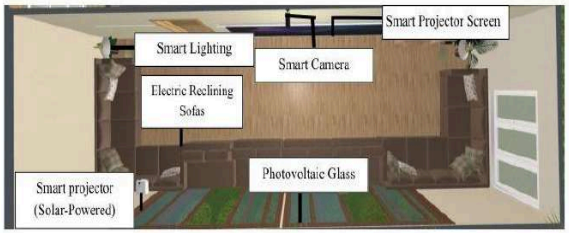
Smart Warden Room Interior:



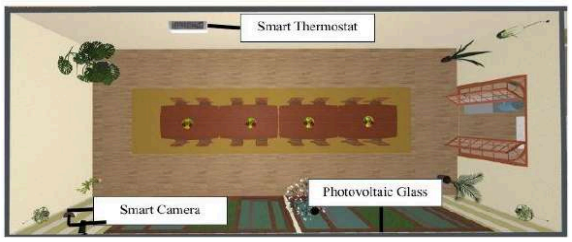
Smart Kitchen Interior:



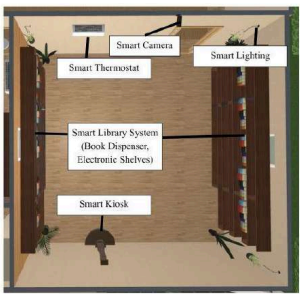
Smart Entertainment Room:



Smart Living Room:



Smart Library:



3D Interior Design:

3D Exterior Design:

6.1.3) Bedroom and Assistive Sleep Systems:

The bedroom acts as the main space for continuous health monitoring. A key component is the IoT-based smart mattress, which uses under-mattress piezoelectric sensors to track vital signs and help prevent issues linked to limited movement.

- For pressure ulcer prevention, the system includes pressure mapping and automatic repositioning. It can also notify the warden if a resident remains in the same position for too long, reducing the risk of pressure sores (Kim et al., 2025).
- To support high-risk residents, bed-exit sensors send real-time alerts to the warden's dashboard. This is especially important for individuals with mobility difficulties or those who may wander at night.
- Residents are also able to control their immediate environment using an NFC or voice-enabled smart apartment system. This allows hands-free adjustment of features such as lighting and curtains, supporting comfort and reinforcing a sense of independence.

6.1.4) Health Monitoring Systems

The smart home integrates health monitoring technologies to support continuous supervision of elderly residents. Smart medication dispensers provide automated dosage control, reducing errors. Vital monitoring systems track heart rate, blood pressure, and sleep patterns via sensors, with data sent in real time to a warden dashboard. Automated alerts are triggered for abnormal readings, enabling timely medical intervention and improving overall safety.

6.1.5) Smart Library and Cognitive Engagement Centre

The library is designed to support cognitive health and reduce social isolation through accessible and easy-to-use features.

- Lighting follows circadian rhythm principles, with reduced blue light and warmer tones in the evening. This helps improve sleep quality and can reduce symptoms of depression and agitation (Wang et al., 2024).

- Assistive reading is supported through smart magnification tables and AI-enabled displays, which enlarge text and adjust contrast for residents with visual impairments, supporting independent reading (Rodriguez et al., 2026).
- Seating includes bone-conduction audio systems, allowing residents to listen to audiobooks clearly without headphones or disturbing others.
- Smart book dispensing shelves enable easy access to books with minimal physical effort, while interactive smart kiosks help residents search for materials and access digital content independently.

6.1.6) Integrated Safety and Emergency Response Systems

Safety across the facility is managed through a coordinated network of sensors and communication systems designed for fast response.

AI-enabled cameras are used in communal areas to detect falls and unusual behaviour. Falls remain one of the leading causes of injury-related deaths worldwide, and the use of behavioural monitoring allow for faster response compared to manual supervision.

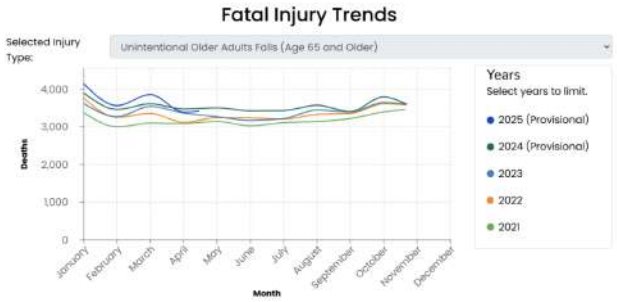


Figure 12 : Falls are a leading cause of injury and death among adults ages 65+ (CDC WISQARS, 2024)

Emergency systems, including panic buttons and smoke or carbon monoxide detectors, operate using the MQTT protocol. This ensures rapid communication with very low delay, which is critical in emergency situations where response time directly affects outcomes (Seth et al., 2025).

For residents with cognitive conditions such as dementia, additional safeguards are in place. These include geo-fenced GPS tracking and NFC-based access control to reduce the risk of unsafe wandering, while still maintaining a non-restrictive environment for other residents.

6.1.7) Smart Energy and Environmental Sustainability

The facility integrates renewable energy systems to improve efficiency and reduce long-term operational costs.

- Bifacial solar panels are used to capture sunlight from both sides, increasing energy generation under different environmental conditions. This improves overall system performance compared to conventional panels.
- An AI-based energy forecasting system predicts solar output using short-term weather data. Based on these predictions, energy-intensive tasks such as laundry and kitchen operations are scheduled during peak generation periods to make better use of available energy.
- Outdoor areas are supported through solar-powered smart gazebos equipped with adjustable louvers. These provide shaded, comfortable spaces for residents while also contributing to the facility's energy ecosystem.
- In addition, the use of transparent photovoltaic glass in selected areas, such as the library, helps reduce glare while generating energy. This supports both user comfort and sustainability goals.

6.1.8) Assistive Smart Living Devices

Assistive technologies enhance comfort, independence, and safety. Voice assistants such as Amazon Alexa enable hands-free control of home functions. Smart beds offer adjustable positioning and fall alerts, while kitchen shut-off systems reduce fire risks. Automated curtains and blinds further improve accessibility and convenience.

6.1.9) Application Interface

The application interface for the elderly home is designed to prioritise safety, health monitoring, and ease of use. It provides wardens with a centralized dashboard for real-time monitoring of residents and quick response to emergencies.

Key features include:

- **Health Monitoring Dashboard:** Displays vital signs, sleep patterns, and alerts for abnormal readings.
- **Fall Detection and Emergency Alerts:** Sends instant notifications to wardens for rapid intervention.
- **Simplified User Interface:** Voice-enabled controls and minimal interaction for elderly residents.
- **Central Monitoring System:** Allows wardens to oversee multiple residents from a single platform

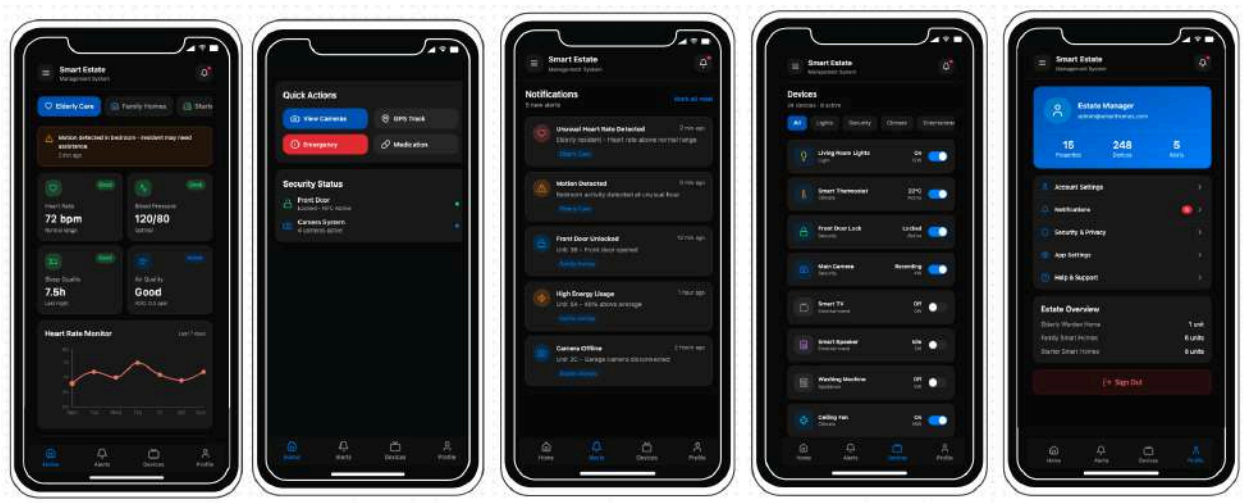


Figure 13 : Smart Guard Application interface for Warden home

6.2) 6x 4 bedroom houses

6.2.1) Overview:

The 4-bedroom family house achieves the right balance between convenience, energy efficiency and safety in order to accommodate modern households with kids. The home consists of

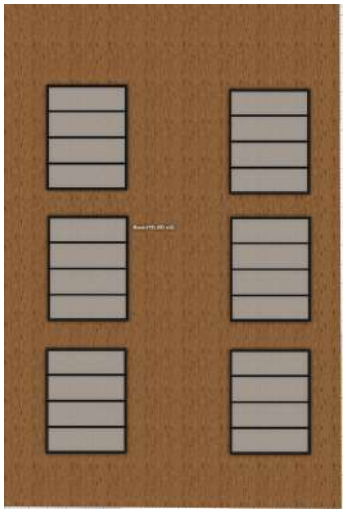
- 1 master bedroom (parents)
- 2 children bedroom
- 1 study home/office room

2D Design Overview

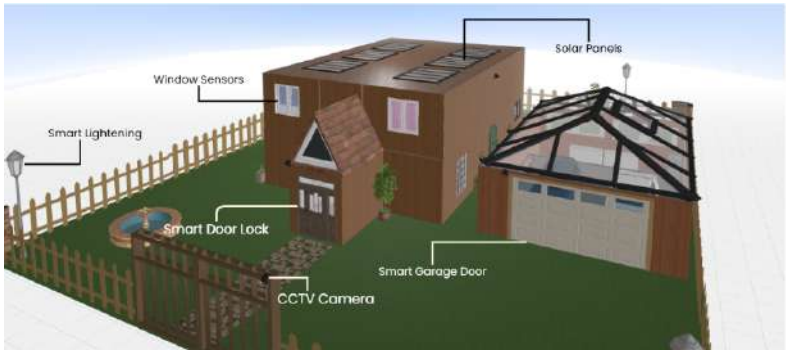
Ground floor

1st Floor

Roof View



Smart Home Exterior Design:



Every room's unique requirements are taken into consideration while designing the smart home system, guaranteeing a secure and effective living space for every resident.

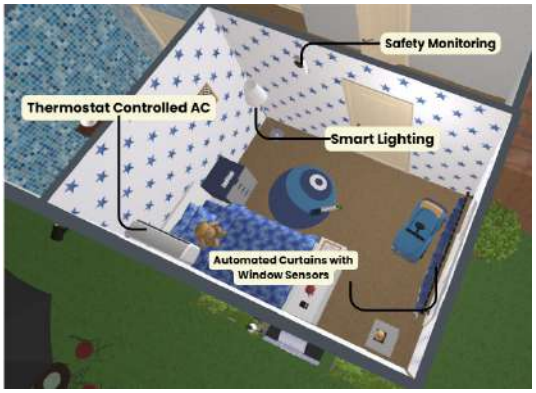
6.2.2) *Room Based Smart Design*

Parents Bedroom:

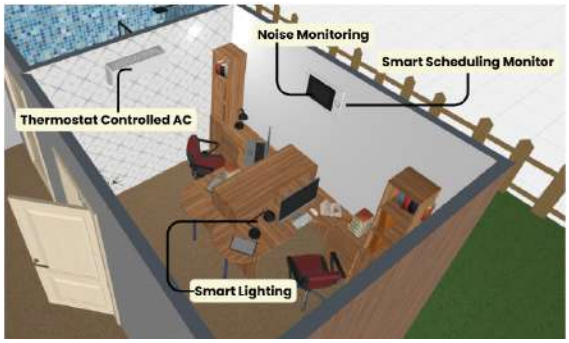
- Smart lightning for dimming control
- Voice assistant such as Alexa
- Smart thermostat for temperature adjustments

Childrens Bedroom

- Screen time control to manage device usage
- Smart child safety monitoring using motion sensors with behavioral AI
- Smart lightning with an automatic night mode



Study/office room

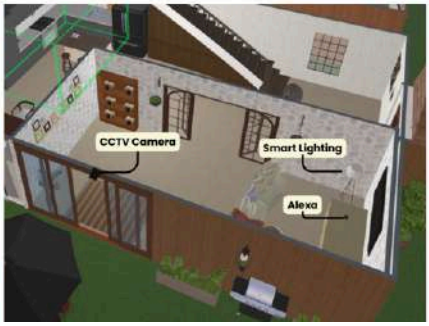


- Controlled internet access for study purposes
- Smart lightning for ideal working environments
- Smart scheduling system for reminders and task notifications
- Noise monitoring wherever loud noises are recognized, a notification is sent right away.

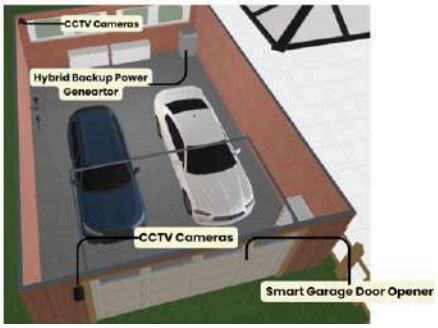
Other:

Smart Kitchen

Living Room



Garage



Hallways

6.2.3) *Safety and Security Systems:*

Particularly in a family home with kids, safety and security are crucial elements of smart home design. The proposed system incorporates modern technology in order to keep tabs on activities, avoid accidents and react fast to emergencies while guaranteeing a safe living environment for all residents.
The following smart safety features are applied:

- **Smart child safety monitoring system (AI + Motion sensors):**
  This system tracks movement in the house using motion sensors and behavioral AI. It can identify odd behavior, such kids entering forbidden areas or abruptly departing the house. Parents receive real-time warnings through the mobile application when such events take place.
- **Smart Kitchen Safety:**
  Safety sensors are applied throughout the kitchen in order to detect any smoke, gas leaks and overheating equipment. By automatically shutting off appliances and alerting residents, the technology minimizes the possibility of accidents or fires.
- **AI enabled smart camera for fall detection:**
  To keep an eye on activities, smart cameras are placed in strategic locations throughout the house. Abnormal behavior, such as falls or extended periods of inactivity, can be detected by the system. Alerts are automatically transmitted to the homeowner or the appropriate authorities in an emergency, allowing for quick action.

6.2.4) Application Interface

The application interface for family homes focuses on convenience, security, and parental control. It enables homeowners to manage household systems efficiently while ensuring a safe environment for all residents.
Key features include:

- **Security Monitoring:** Real-time CCTV access and motion-based alerts.
- **Parental Control Features:** Screen time management and activity monitoring for children.
- **Smart Device Control:** Remote control of lighting, appliances, and thermostats.
- **Energy Monitoring:** Tracks energy usage and optimises consumption using solar integration

Figure 14 : Smart Guard Application interface for 6 x 4 bedroom home

6.3) 8x 2 bedroom houses

6.3.1) Overview and rational

The 2-bedroom smart homes are designed as affordable starter homes, primarily targeting young professionals and small families. The focus of the design is to provide essential smart home features that enhance security, convenience, and energy efficiency, while maintaining a cost-effective solution.
Each unit consists of:

- Master bedroom
- Smart gym room
- Living area
- Kitchen
- Garage

The inclusion of a smart gym room reflects modern lifestyle trends, supporting health and wellness within the home environment. This design increases property value and usability while remaining aligned with the needs of the target occupants

6.3.2) Layout and design

Roof view

Smart Home Exterior Design:

Master bedroom

Smart lighting with automation and dimming

- Smart thermostat for temperature control
- Voice assistant integration
- Automated curtains

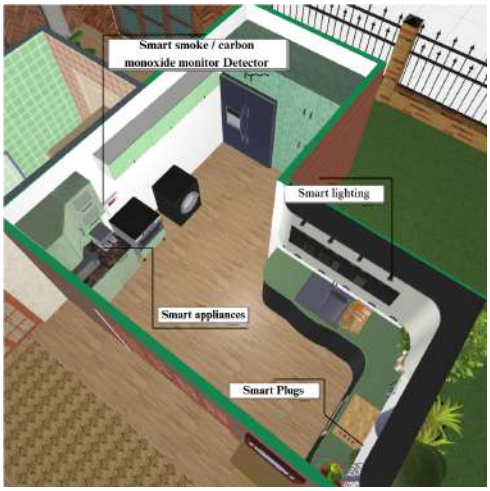
This provides personalised and comfortable living environment

Smart Gym

Smart mirror providing real-time feedback and guidance using AI motion tracking system to monitor posture, workouts, and movement patters

- Behavioral AI with fall detection for safety

Kitchen

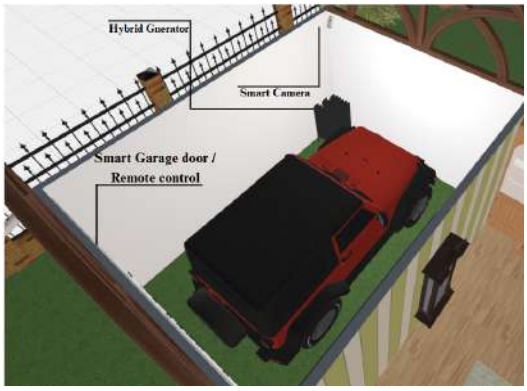


Smart Appliance control via mobile application

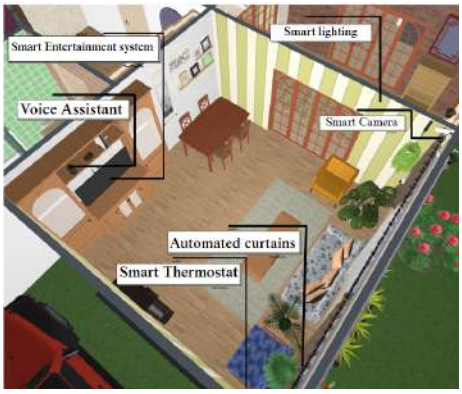
- Smart smoke and gas detection systems
- Automatic shut off mechanisms
- Smart plugs for energy management

Enhances safety and efficiency in daily activities

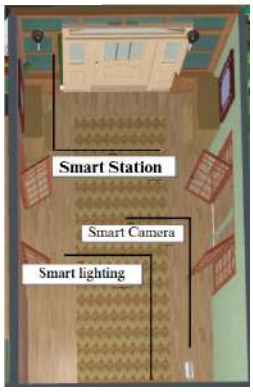
Garage



Living room



Hallway



Smart Living and Access Control System :

- Smart automated lighting with motion detection for energy efficiency and visibility
- AI-enabled CCTV monitoring for continuous security surveillance
- Smart monitoring system in the hallway for real-time activity tracking
- Motion sensors that trigger alerts for unusual activity
- Voice assistant for centralized control of devices and automation
- Smart entertainment system supporting user interaction

Garage Integration:

- Smart garage door with remote control via mobile application
- Vehicle entry and exit logging system for tracking activity
- Motion detection and CCTV monitoring for enhanced security
- Full integration with mobile application for real-time monitoring and control

Overall Benefits:

- Acts as a central hub for user interaction and automation

- Improves security across hallway and garage areas
- Enables real-time monitoring, alerts, and remote access
- Enhances convenience, control, and system efficiency

6.3.3) Application Interface

The application interface for starter homes is designed to provide essential smart home functionality in a cost-effective and user-friendly manner. It focuses on security, automation, and lifestyle enhancement.
Key features include:

- Centralised Control: Manage lighting, appliances, and smart locks through a single interface.
- Smart Security System: AI-enabled CCTV, motion detection, and real-time alerts.
- Energy Management: Monitor electricity usage and optimize solar energy consumption.
- Lifestyle Integration: Supports smart gym tracking and daily activity monitoring.

Figure 15 : Smart Guard Application interface for 8x 2 bedroom home

6.3.4) Safety and Security systems

- AI-enabled CCTV with behavioural detection.Abnormal behavior, such as falls or extended periods of inactivity, can be detected by the system. Alerts are automatically transmitted to the homeowner or the appropriate authorities in an emergency, allowing for quick action.
- Smart locks (PIN, biometric, or app-controlled)
- Motion detection systems to detect intrusion
- Smart gate automation with remote control
- Smart Kitchen Safety in which sensors are throughout the kitchen in order to detect any smoke, gas leaks and overheating equipment. By automatically shutting off appliances and alerting residents, the technology minimizes the possibility of accidents or fires.
- Centralised mobile application for:
    - Security control
    - Appliance management
    - Real-time alerts

These systems provide intelligent monitoring and remote accessibility, enhancing overall home security.

6.3.5) Smart Energy and Power Systems

- Solar panels (standard across all homes)
- Smart energy monitoring systems
- Automated appliance usage during peak solar generation
- Hybrid backup system (battery + generator)

Smart energy systems contribute to reduced energy consumption and improved sustainability

7) Benefits

This section outlines both the tangible and intangible benefits of the proposed smart home system, highlighting its financial value as well as its impact on user experience and quality of life.Following are the tangible and intangible benefits

Tangible benefits	Intangible benefits
Reduced electricity costs via solar panels + smart energy monitoring (Section 7.3.9 → supports ROI in Section 8.2)	Improved convenience through mobile app + voice assistant integration (centralised control system)
Lower energy consumption via automated lighting, smart thermostats, and smart plugs	Enhanced security and peace of mind via real-time monitoring, alerts, and CCTV
Optimised energy usage during peak solar generation	Improved safety using motion detection, smart locks, and AI surveillance
Reduced maintenance costs with real-time fault detection	Seamless user experience through a centralised smart home ecosystem (Section 7.3.9)
Reduced maintenance costs through real-time fault detection and alerts	Better lifestyle support through smart gym and health-focused features
Increased property value due to modern smart home features (AI CCTV, automation, smart gym)	Increased comfort with automated lighting, temperature, and environment control
Lower security costs using AI-enabled CCTV and smart locks instead of manual systems	Time-saving through automation of daily tasks (garage, lighting, appliances)
Reduced insurance risk due to enhanced safety and surveillance systems	Seamless user experience through a centralised smart home ecosystem (Section 7.3.9)
Improved operational efficiency through integrated mobile application control	Modern living aligned with UAE smart city and sustainability vision
Higher rental/resale potential for tech-enabled starter homes targeting young professionals	Increased user satisfaction and quality of life for young professionals and small families

Table 9 : Tangible and intangible benefits

8) COST ANALYSIS

This section presents the cost analysis of the proposed smart home systems, evaluating the investment required and the financial feasibility of each housing type.

8.1) Bills of Quantities (BOQ)

SMART HOMES COST ANALYSIS

Warden Home		6 x 4 Bedroom House		8 x 2 Bedroom House	
Product Name	Cost (AED)	Product Name	Cost(AED)	Product Name	Cost (AED)
8MP HikVision Camera DS-2CD2186G2-I (SU) x 22	29700.00	8MP HikVision Camera DS-2CD2186G2-I (SU) x 7	9450.00	8MP HikVision Camera DS-2CD2186G2-I (SU) x 6	8100.00
Sharaf DG Smart Apartment Bundle *(19 x Smart Switches / 2 x Smart Thermostat/ 18 x Curtain Bundle/ 3 x smart door lock/ Smart Station)*	25149.00	Sharaf DG Smart Apartment Bundle *(7 x Smart Switches/ 2 x Smart Thermostat/ 2 x Curtain Bundle/ 1 x smart door lock / Smart Station)*	11999.00	Sharaf DG Smart Apartment Bundle *(7 x Smart Switches/ 2 x Smart Thermostat/ 2 x Curtain Bundle/ 1 x smart door lock / Smart Station)*	11999.00
PHILIPS Bulb x 55	467.50	EverBright Motion Sensor (6 per pack)	168.00	EverBright Motion Sensor (6 per pack)	168.00
Voice Assistant (Alexa voice assistant) x 15	3450.00	PHILIPS Bulb x 36	306.00	PHILIPS Bulb x 35	300.00
MIO 430S+ SMART x 14	338898.00	Voice Assistant (Alexa voice assistant) x 2	460.00	Voice Assistant (Alexa voice assistant) x 2	460.00
EverBright Motion Sensor (6 per pack)	168.00	SR Series Wireless Panic Button 60-A100-00PB-SR-50-11 (White) x 2	1798.00	SC5 Battery Smart Smoke and Carbon Monoxide Alarm x 2	947.68
SR Series Wireless Panic Button 60-A100-00PB-SR-50-11 (White) x 14	12586.00	SC5 Battery Smart Smoke and Carbon Monoxide Alarm x 2	947.68	Ring B09D38X4Z3 2nd Gen Alarm x 2	1698.00
SC5 Battery Smart Smoke and Carbon Monoxide Alarm x 3	1421.52	Ring B09D38X4Z3 2nd Gen Alarm x 3	2547.00	EMPORIA ENERGY Vue 3 Commercial 3-Phase Smart Monitor	809.81
Ring B09D38X4Z3 2nd Gen Alarm x 20	16980.00	EMPORIA ENERGY Vue 3 Commercial 3-Phase Smart Monitor	809.81	EcoFlow Smart Generator 4000 (Dual Fuel)	5000.00
EMPORIA ENERGY Vue 3 Commercial 3-Phase Smart Monitor x 2	1619.62	EcoFlow Smart Generator 4000 (Dual Fuel)	5000.00		
EcoFlow Smart Generator 4000 (Dual Fuel)	10000.00	Minut - Noise, Occupancy, Motion and Temperature Monitoring	17.00		
Minut - Noise, Occupancy, Motion and Temperature Monitoring	17.00				
Austell 5-Seater Fabric Electric Recliner	9000.00			Total	29482.49
Total	449456.64	Total	33502.49		

Smart Home Complete Project Total	Cost
Warden Home	449456.64
6 x 4 Family Bedroom House	201014.94
8 x 2 Youth Bedroom House	235859.92
Final Total	886331.50

8.1.1) Warden home

Product	**Unit**	**Rate (AED)**	**Amount (AED)**	Link
8MP HikVision Camera DS-2CD2186G2-I (SU)	*22*	*1350.00*	*29700*	https://hikvisioncameradubai.com/product/hikvision-ds-2cd2186g2-isu/
Sharaf DG Smart Apartment Bundle (19 x Smart Switches / 2 x Smart Thermostat/ 18 x Curtain Bundle/ 3 x smart door lock/ Smart Station)	*1*	*11,999.00*	*25149.00*	https://uae.sharafdg.com/product/aquara-smart-comfort-kit-9-x-smart-switches-and-motion-sensor-3-x-smart-thermostat-with-necessary-conversion-kit-3-x-curtain-control-bundle/?dg=false
PHILIPS Bulb	*55*	*8.50*	*467.05*	https://amzn.eu/d/0hzm7NMS
Voice Assistant (Alexa voice assistant)	*15*	*229.99*	*3450.00*	https://amzn.eu/d/0fNRks2N
MIO 430S+ SMART	*14*	*24207.00*	*338898.00*	https://restonic.ae/product/mio-430s-smart-adjustable-bed-restonic
EverBright Motion Sensor (6 per pack)	*2*	*84.99*	*168.00*	https://amzn.eu/d/0dbtrNhY
SC5 Battery Smart Smoke and Carbon Monoxide Alarm	*3*	*429.07*	*1287.21*	https://amzn.eu/d/0hldUb7C
Ring B09D38X4Z3 2nd Gen Alarm	*20*		*16980*	https://uae.sharafdg.com/product/ring-b09d38x4z3-2nd-gen-alarm-security-kit-5pcs-set/?cpn=22461014614&pl=&kw=&adg=&ch=x&utm_source=google&utm_medium=cpc&utm_campaign=LS_BAU_Pmax_Low-Performing_Bilal_180425&utm_term=&device=c&utm_matchtype=&utm_network=x&utm_placement=&utm_content=&utm_Adposition=&utm_location=9047580&utm_Sitelink=}&utm_Audience=&utm_c

				ampaignid=22461014614&utm_adgroupid=&gad_source=4&gad_campaignid=22451081640&gbraid=0AAAAADp8qLaWqaae6YOBIiPHnaps9-FIZ&gclid=CjwKCAjwnN3OBhA8EiwAfpTYeuP2syM3OWTIvqjreny2r2CMcjUaCbB5w97hLxLdxS-CBtc1Pfn0bRoCYVMQAvD_BwE
EMPORIA ENERGY Vue 3 Commercial 3-Phase Smart Monitor	*2*	*1141.36*	*1619.62*	https://amzn.eu/d/06HNGSUy
EcoFlow Smart Generator 4000 (Dual Fuel)	*2*	*8089.00*	*16178.00*	https://desertcart.ae/products/647632030
Minut - Noise, Occupancy, Motion and Temperature Monitoring	*1*	*17.00*	*17.00*	https://desertcart.ae/products/97210332
Austell 5-Seater Fabric Electric Recliner	*1*	*9000.00*	*9000.00*	https://homekode.com/products/ilva-modular-u-shaped-sofa

Table 10: Warden Home BOQ

8.1.2) 6x 4 bedroom home

Product	**Unit**	**Rate (AED)**	**Amount (AED)**	Link
8MP HikVision Camera DS-2CD2186G2-I (SU)	*7*	*1350.00*	*9450.00*	https://hikvisioncameradubai.com/product/hikvision-ds-2cd2186g2-isu/
Sharaf DG Smart Apartment Bundle (7 x Smart Switches/ 2 x Smart Thermostat/ 2 x Curtain Bundle/ 1 x smart door lock / Smart Station)	*1*	*11,999.00*	*11,999*	https://uae.sharafdg.com/product/aquara-smart-comfort-kit-9-x-smart-switches-and-motion-sensor-3-x-smart-thermostat-with-necessary-conversion-kit-3-x-curtain-control-bundle/?dg=false
PHILIPS Bulb	*36*	*8.50*	*306.00*	https://amzn.eu/d/0hzm7NMS
Voice Assistant (Alexa voice assistant)	*2*	*229.99*	*459.98*	hthttps://amzn.eu/d/0fNRks2N
EverBright Motion Sensor (6 per pack)	*2*	*84.99*	*168.00*	https://amzn.eu/d/0dbtrNhY
SC5 Battery Smart Smoke and Carbon Monoxide Alarm	*2*	*429.07*	*858.14*	https://amzn.eu/d/0hldUb7C
Ring B09D38X4Z3 2nd Gen Alarm	*3*	*849.00*	*2547.00*	https://uae.sharafdg.com/product/ring-b09d38x4z3-2nd-gen-alarm-security-kit-5pcs-set/?cpn=22461014614&pl=&kw=&adg=&ch=x&utm_source=google&utm_medium=cpc&utm_campaign=LS_BAU_Pmax_Low-Performing_Bilal_180425&utm_term=&device=c&utm_matchtype=&utm_network=x&utm_placement=&utm_content=&utm_Adposition=&utm_location=9047580&utm_Sitelink=}&utm_Audience=&utm_campaignid=22461014614&utm_adgroupid=&gad_source=4&gad_campaignid=22451081640&gbraid=0AAAAADp8qLaWqaae6YOBIiPHnaps9-FIZ&gclid=CjwKCAjwnN3OBhA8EiwAfpTYeuP2syM3OWTIvqjreny2r2CMcjUaCbB5w97hLxLdxS-CBtc1Pfn0bRoCYVMQAvD_BwE
EMPORIA ENERGY Vue 3 Commercial 3-Phase Smart Monitor	*1*	*1141.36*	*1141.36*	https://desertcart.ae/products/647632030
EcoFlow Smart Generator 4000 (Dual Fuel)	*1*	*8089.00*	*8089.00*	https://desertcart.ae/products/97210332
Minut - Noise, Occupancy, Motion and Temperature Monitoring	*1*	*17.00*	*17.00*	https://desertcart.ae/products/97210332

Table 11 : 6x 4 family home BOQ

8.1.3) 8x 2 Bedroom Home

Product	Unit	Rate (AED)	Amount (AED)	Link
8MP HikVision Camera DS-2CD2186G2-I (SU)	*6*	*1350.00*	*8100.00*	https://hikvisioncameradubai.com/product/hikvision-ds-2cd2186g2-isu/
Sharaf DG Smart Apartment Bundle (7 x Smart Switches/ 2 x Smart Thermostat/ 2 x Curtain Bundle/ 1 x smart door lock / Smart Station)	*1*	*11,999.00*	*11999.00*	https://uae.sharafdg.com/product/aquara-smart-comfort-kit-9-x-smart-switches-and-motion-sensor-3-x-smart-thermostat-with-necessary-conversion-kit-3-x-curtain-control-bundle/?dg=false
PHILIPS Bulb	*36*	*8.50*	*306.00*	https://amzn.eu/d/09XFxRgo
Voice Assistant (Alexa voice assistant)	*2*	*229.99*	*306.00*	https://amzn.eu/d/0d0otr9v
EverBright Motion Sensor (6 per pak)	*2*	*84.99*	*168.00*	https://amzn.eu/d/0d9Gpl9t
SC5 Battery Smart Smoke and Carbon Monoxide Alarm	*2*	*429.07*	*858.141*	https://amzn.eu/d/028Ru7Vi
Ring B09D38X4Z3 2nd Gen Alarm	*2*	*849.00*	*1698.00*	https://uae.sharafdg.com/product/ring-b09d38x4z3-2nd-gen-alarm-security-kit-5pcs-set/?cpn=22461014614&pl=&kw=&adg=&ch=x&utm_source=google&utm_medium=cpc&utm_campaign=LS_BAU_Pmax_Low-Performing_Bilal_180425&utm_term=&device=c&utm_matchtype=&utm_network=x&utm_placement=&utm_content=&utm_Adposition=&utm_location=9047580&utm_Sitelink=}&utm_Audience=&utm_campaignid=22461014614&utm_adgroupid=&gad_source=4&gad_campaignid=22451081640&gbraid=0AAAAADp8qLaWqaae6YOBIiPHnaps9-FIZ&gclid=CjwKCAjwnN3OBhA8EiwAfpTYeuP2syM3OWTIvqjreny2r2CMcjUaCbB5w97hLxLdxS-CBtc1Pfn0bRoCYVMQAvD_BwE
EMPORIA ENERGY Vue 3 Commercial 3-Phase Smart Monitor	*1*	*1141.36*	*1141.36*	https://amzn.eu/d/01mp3H4C
EcoFlow Smart Generator 4000 (Dual Fuel)	*1*	*8089.00*	*8089.00*	https://desertcart.ae/products/647632030

Table 12: Starter Home BOQ

8.2) Return On investment (ROI)

The Return on Investment (ROI) analysis evaluates the financial and operational benefits of implementing smart home technologies across the residential estate. The primary financial return is achieved through energy savings, automation, and operational efficiency improvements.Smart home systems are known to reduce energy consumption through automation and monitoring, contributing to long-term cost savings

(Jain, 2025).

Housing Type	Annual Savings AED	Total System Cost	Per estate cost	ROI (%)	Payback period
Warden Home	25,000	449,456.64	N/A	5.67%	18 years
6 x 4 Bedroom House	12,000 (2000 x 6)	201,014.94	33,502.49	5.97%	17 years
8 x 2 Bedroom House	12,000 (1500 x 8)	235,859.92	29,482.49	5.09%	20 years

Table 13 : Return on investment and payback period

Detailed calculations are illustrated in ***Appendix A***.

9) ADDITIONAL FEATURES (EXTRA COST)

A variety of sophisticated and optional features can be added to the basic smart home systems to improve comfort, convenience, and lifestyle in all three types of properties. These features are available as upgrades based on user requirements and financial constraints and are not part of the base cost.

ADDITIONAL FEATURES COST ANALYSIS		
Product Name	Cost (AED)	Links
Yardian Pro Smart Sprinkler Controller 8 Zone	969	https://amzn.eu/d/02P7E4JV
AEKE Smart Home Gym K1	13213.65	https://aeke.com/products/aeke-home-gym-k1?variant=51274362257572&country=AE&currency=USD&utm_medium=product_sync&utm_source=google&utm_content=sag_organic&utm_campaign=sag_organic&srsltid=AfmBOoojqO80liJcIGm40mfTwO-PBB1RC-puUQobeqUDGc7d51wdXO9zEa0
VINDSTYRKA Air quality sensor, smart	129	https://www.ikea.com/ae/en/p/vindstyrka-air-quality-sensor-smart-00498231/?utm_source=google&utm_medium=organic-shopping&utm_campaign=shopping_feed&utm_content=free_google_shopping_clicksLightingHomeelectronics
Genie 7155-TKV Smart Garage Door Opener StealthDrive Connect	2397	https://www.desertcart.ae/products/409110142-genie-7155-tkv-smart-garage-door-opener-stealthdrive-connect-ultra
Apple Watch Ultra	3199	https://www.apple.com/ae/apple-watch-ultra-3/
KLIM Smart Pill Dispenser Machine	515	https://www.desertcart.ae/products/763389093-klim-smart-pill-dispenser-machine-with-alarm-wifi-app-monitoring
Smart Library Bookshelf Book File Racks Mobile Shelving Storage System Mobile File Compactor Automatic Archive Cabinet	694.1	https://www.made-in-china.com/price/prodetail_Library-Furniture_sTarUnXcgjVI.html?utm_source=google&utm_medium=shopping&utm_campaign=sTarUnXcgjVI

Library Self Checkout Kiosk Touch Screen Terminal with RFID Scanner for Book Borrowing and Return System	2983	https://yataer.en.made-in-china.com/product/jRcrGyLvnNVT/China-Library-Self-Checkout-Kiosk-Touch-Screen-Terminal-with-RFID-Scanner-for-Book-Borrowing-and-Return-System.html?utm_source=google&utm_medium=shopping&utm_campaign=2285311885
Nourest Orion 3D Rocking Recline Massage chair With Ottoman V2.0 | Beige	3510	https://activefitnessstore.com/nourest-orion-3d-rocking-recline-massage-chair-with-ottoman-v2.0-beige-orion-v2-beige?gad_source=4&gad_campaignid=21117278635&gbraid=0AAAAACPgCpJ6HlFlYeDm23dPmnW0FUn3_&gclid=CjwKCAjwnN3OBhA8EiwAfpTYevlgRRWgZZPqGEa1X6fbeL3ipEvlH1IekkDL4IU9oUI4l89thKh6phoCrkMQAvD_BwE
Rabbit R1 AI Personal Assistant	859	https://www.noon.com/uae-en/rabbit-r1-ai-personal-assistant-smart-voice-activated-home-hub-with-built-in-display-music-streaming-smart-device-control-and-virtual-scheduling/ZB6278D73F3471F9DA120Z/p/?o=f70f7c4e2ae4794f&shareId=ffe55ae3-d7c4-4f67-9195-97ae3dd95c11
Nebula Solar Portable Projector	2809	https://ae.tejar.com/nebula-solar-portable-projecto
Powerology 100" Motorised Projector Screen	1499	https://uae.microless.com/product/powerology-100-motorised-projector-screen-169-widescreen-viewing-wireless-remote-control-90-150-diagonal-screen-easy-installation-for-indoor-outdoor-white/
Total	32776.75	

Table 14: Additional Features

10) NON-NEGOTIABLES

All three categories of properties - family houses, senior housing, and starter homes—must incorporate some fundamental smart home components. All occupants are guaranteed a sufficient level of safety, security, functionality, and energy efficiency due to these non-negotiable elements.

- **Security Systems:** Every property will include basic security outlines such as smart cameras, motion sensors and smoke/carbon monoxide detectors.

- **Smart Home Platform Integrations:** To guarantee consistent control and integration of all devices, a central smart home platform (such a voice assistant system) must be installed.
- **Energy Monitoring and Solar Systems:** Every property must include solar panel integration and energy monitoring systems to track energy usage and optimise efficiency.

- **Basic Automation and Remote Access:** Each property must use a basic home automation system like smart lighting control and mobile application-based remote access.

- **Secure Network and Data Protection:** Strong security measures such as encrypted communications, user authentication and secure wifi networks are essential to ensure protection against cyber threats or unauthorized access.

- **Backup Power System Backup:** To guarantee continuous operation during outages, critical systems like security and monitoring must have backup power support.

11) CONCLUSION

The proposed smart housing estate effectively integrates modern smart technologies across all three residential categories: 4-bedroom family homes, elderly accommodation, and 2-bedroom starter homes. Each housing type has been carefully designed to address the specific needs of its occupants, with family homes prioritising safety and convenience, elderly accommodation focusing on accessibility and health monitoring, and starter homes providing cost-effective and efficient smart living solutions.

The project demonstrates how the integration of IoT systems, automation technologies, and energy-efficient solutions, such as solar power and smart energy management, can create a residential environment that is both sustainable and technologically advanced. In addition, the inclusion of security systems, real-time monitoring, and user-friendly application interfaces enhances both safety and usability across the estate.

From a practical perspective, the incorporation of cost analysis and return on investment (ROI) further strengthens the feasibility of the proposed solutions, ensuring that the designs are not only innovative but also economically viable. This highlights the importance of balancing technological advancement with real-world constraints such as cost, scalability, and system compatibility.

Overall, the project presents a scalable and adaptable smart home framework that aligns with the growing demand for intelligent residential environments. As smart technologies continue to evolve, such integrated systems have the potential to play a key role in shaping future housing developments, supporting sustainability goals, improving quality of life, and enabling more efficient use of resources.

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13) Appendix A

ROI Calculations:

Warden Home

- *Total System Cost = 449,456.64 AED*
- *Annual Savings = 25,000 AED*

ROI:

ROI = (25000 / 449456.64)×100
ROI = 0.0556 x 100 = 5.56 %

Payback Period:

Payback = 449456.64 / 25000 =17.98
= 18 yearsPayback

6x 4 Bedroom home

- System Cost = **201,014.94 AED**
- Annual Savings per house = **2,000 AED**
- Total houses = 6

Total Annual Savings = 2000 × 6 = 12,000 AED

ROI:

ROI = (12000 / 201014.94) × 100
ROI = 0.0597 × 100 = 5.97%

Payback Period:

Payback = 201014.941 / 2000 = 16.75
= 17 years

8x 2 Bedroom home

- System Cost = **235859.92 AED**
- Annual savings per house = **1500 AED**
- Total houses = 8

Total Annual Savings = 1500 x 8 = 12000

ROI :
(12,000 ÷ 235,859.92) × 100 = 5.09%

Payback period :
235,859.92 ÷ 12,000 = 19.65 = 20 years

14) Appendix B : Relevant Statistics

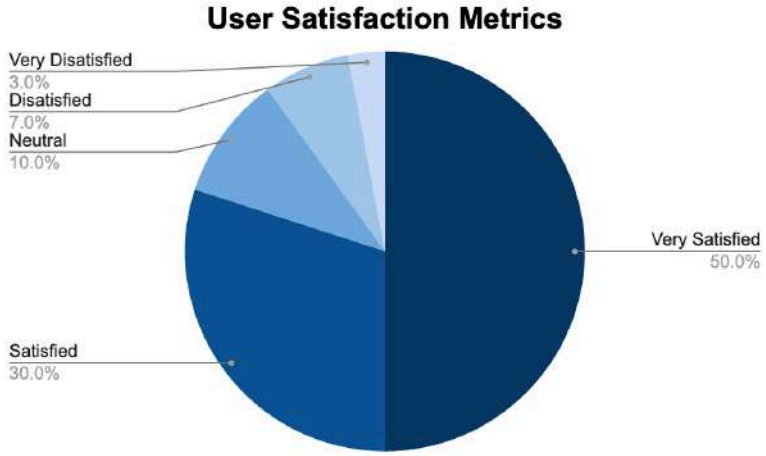

Figure 16 : Tenant Satisfaction Levels in IoT-Enabled Residential and Commercial Properties (%) (World Journal of Advanced Research and Reviews, 2025)

Cost Component	Typical Range (Dubai)	What It Covers	What Clients Should Know
Smart design & coordination	AED 15,000 – 40,000	Automation planning, MEP coordination, load planning	Skipping this often causes higher costs later
Basic home automation	AED 30,000 – 70,000	Lighting, basic AC control, curtains	Suitable for most villas
Advanced automation	AED 80,000 – 150,000+	Climate zoning, energy monitoring, security	Must be planned early with MEP
Smart MEP integration	AED 20,000 – 60,000	Sensors, dashboards, system integration	Reduces long-term energy bills
Annual maintenance	AED 3,000 – 8,000	Updates, system checks	Depends on brand and system complexity

Figure 17 : Approximate cost of housing based on technology (AAA Dubai, 2026)

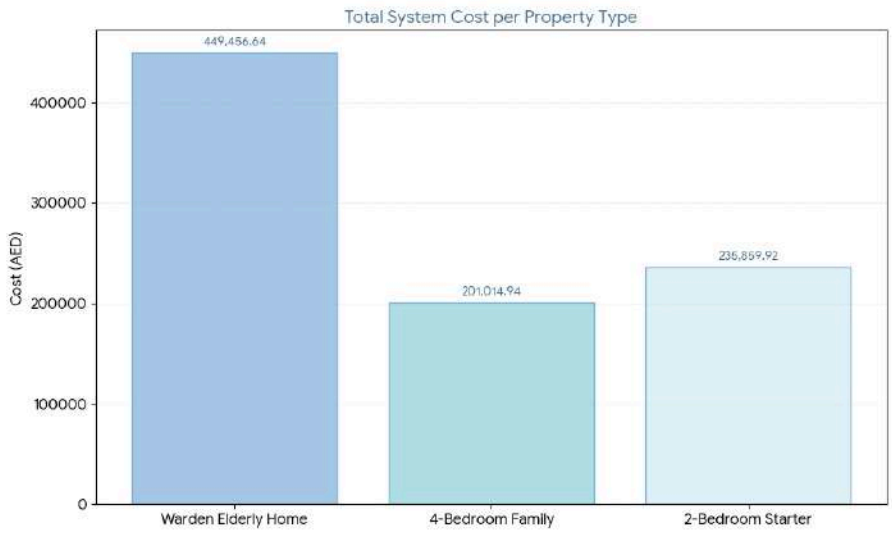

Figure 18: System Capital Expenditure by Property Type

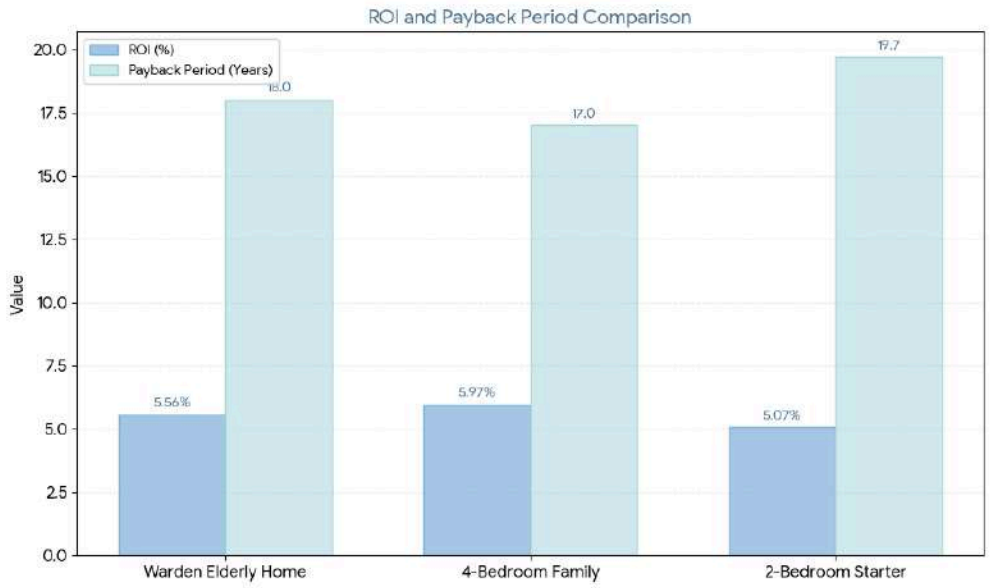

Figure 19 :Comparative Financial Performance: ROI vs. Amortization Period

15) Appendix C : Minutes of the meeting

MINUTES OF THE MEETING

Team Members : 1.Zainab Arslan Dar
2.Rabia Bint Abdul Ahad
3.Raghad Barmem

WEEK 1 :

MEETING 1:

Date: 14/03/2026 **Time:** 2:30-3:30 pm **Attendees:** Raghad, Rabia, Zainab **Session:** Group

Meeting Objective:

- Understand the project requirements
- Brainstorm ideas and develop an initial project plan
- Identify stakeholders and their needs
- Plan project management documents (WBS, PID, Gantt Chart)
- Develop initial design concepts for all properties

Agenda Items:

- Project overview and requirements
- Task distribution among team members
- Initial design ideas and smart features
- Research areas and technologies to be used

Discussion Summary:

The project involves designing a housing estate consisting of:

- 6 × 4-bedroom homes (assigned to Raghad)
- 8 × 2-bedroom homes (assigned to Rabia)
- A warden/elderly care home for 14 residents (assigned to Zainab)

Key requirements identified:

- Integration of solar panels
- Implementation of smart home technologies
- Creation of a digital model, report, and presentation

Ideas for elderly/warden home:

- RFID smart ring with GPS tracking for secure access
- Smart radiator valves for temperature control
- Medication reminder systems
- AI-based smart cameras for emergency detection
- Proper staff scheduling for efficient care
- Backup generators for power reliability

Ideas for residential homes:

- Advanced authentication systems (retina, RFID, fingerprint)
- Energy generation through footsteps (innovative concept)
- Smart home mobile application for control and monitoring
- Integration with voice assistants (e.g., Alexa)
- Smart delivery box for secure package handling
- Motion sensors for automation and security

Decisions Made:

- Task allocation finalized among team members
- Smart home and sustainability features will be core focus
- Research into available technologies and products will be conducted
- All properties will include energy-efficient and automated systems

Tasks Allocated:

Task	Assigned
Design of 6 × 4-bedroom homes	Raghad
Design of 8 × 2-bedroom homes	Rabia
Design of warden/elderly care home	Zainab
All members: Conduct research on smart technologies and begin planning documents	

Progress Review :

Current project status: On track

<u>Completed:</u>

- Understanding of project requirements
- Task allocation
- Initial brainstorming and idea generation

<u>Pending:</u>

- Detailed designs
- Creation of WBS, PID, and Gantt chart
- Development of digital models

Changes to Plan:

Changes made:

- Inclusion of advanced smart home features and sustainability elements

Reason:

- To enhance innovation, efficiency, and align with modern housing and care standards

Next Steps:

- Begin detailed design of assigned properties
- Develop WBS and project documentation
- Start digital modelling (Planner 5D)
- Continue research on smart technologies and implementation

Meeting Conclusion:

The meeting successfully established a clear understanding of the project requirements, assigned responsibilities, and generated innovative ideas. The team is aligned and ready to proceed with the design and planning phase.

MEETING 2:

Date: 26/03/2026 **Time:** 4:00-5:00 pm **Attendees:** Raghad, Rabia, Zainab, Mukarram **Session:** Seminar

Meeting Objective:

- Finalize and present the Project Initiation Document (PID)
- Present project plan to seminar tutor
- Receive feedback and identify improvements

Agenda Items:

- Review and finalize PID
- Present project plan and ideas to tutor
- Discuss feedback and required changes

Discussion Summary:

The team completed the Project Initiation Document (PID) including:

- Project scope
- Objectives
- Stakeholders
- Initial timelines

The project plan and initial ideas were presented to the seminar tutor
Feedback received from tutor included:

- Need for clearer structure in project planning
- More detailed breakdown in tasks (WBS)
- Ensure realistic and feasible smart home features
- Improve clarity in design concepts

Team discussed how to implement the suggested improvements

Decisions Made:

- Revise PID based on tutor feedback
- Improve WBS with more detailed task breakdown
- Refine smart home features to ensure practicality
- Continue development of Gantt Chart

Tasks Allocated:
Raghad: Update PID and begin refining 4-bedroom design
Rabia: Work on WBS and assist in Gantt Chart improvements
Zainab: Refine elderly home design and safety features, Update PID, Work on WBS, and Gantt Chart.
All members: Apply tutor feedback to project planning

Progress Review :

Current project status: On track

Comparison with previous plan:

- Completed:
    - PID creation
    - Initial project planning
    - Tutor feedback session
- Pending:
    - Revision of PID
    - Finalization of WBS and Gantt Chart
    - Detailed designs

Changes to Plan:

Changes made:

- PID updated for better clarity and structure
- WBS expanded to include more detailed tasks
- Refinement of smart home features

Reason:

- Based on tutor feedback to improve clarity, feasibility, and overall project quality

Next Steps:

- Continue market research
- Finalize platform
- Commence design and layout for estates

Meeting Conclusion:
The meeting was productive as the team received valuable feedback from the tutor. Necessary improvements were identified, and the project plan is now more structured and aligned with expectations.

WEEK 2 :

MEETING 1:

Date: 31/03/2026 **Time:** 7:00 pm **Attendees:** Raghad, Rabia, Zainab, **Session:** Group

Meeting Objective:
- To review progress of estate layout and design
- To finalise the smart home platform
- To initiate application interface design and naming
- To develop early technical framework ideas
- To conduct market research to support smart features
- To identify products for implementation
- To begin drafting cost analysis and BOQ structure

Agenda Items:
- Review of design progress
- Platform selection
- Application interface discussion
- Market and product research
- Cost analysis and BOQ planning

Discussion Summary:
- The design and layout of all housing types were progressing according to schedule.
- The team finalized the smart home platform as Amazon Alexa, and agreed on the application name “Smart Guard.”
- Initial market research was conducted and incorporated into the introduction, with supporting figures added to relevant sections of the report.
- Early ideas for the technical framework and system integration were discussed.
- The structure for cost analysis and BOQ tables was drafted to ensure consistency across all housing types.
- Commencement of standard research

Decisions Made:

- Amazon Alexa was confirmed as the central smart home platform.
- The application interface was officially named Smart Guard.
- The format for BOQ and cost analysis tables was agreed upon.
- Market research findings would be integrated into the report to support design decisions.

Tasks Allocated:

Task	Assigned
Continue report development (design, research, costing)	All Members
Begin structuring reflective diary	All members
Standards	Zainab
Shared technical framework research	All members
Continue research	All members

Progress Review :

- Current project status: on track
- Comparison with previous plan:
    - Completed tasks :
        - Initial understanding of project requirements
        - Early brainstorming and planning
        - Initial research on smart technologies
        - Standards Research
    - Tasks ongoing :
        - Finalisation of platform selection
        - Detailed design and layout
        - Cost analysis and BOQ structure
        - Standards Research

Next Steps:

No major structural changes; minor refinements in task allocation and BOQ formatting

Meeting Conclusion:

To improve organisation and ensure consistency across all report sections

MEETING 2:

Date: 02/04/2026 **Time:** 3:50 pm **Attendees:** Raghad, Rabia, Zainab,Mukarram **Session:** Seminar

Meeting Objective:
To receive feedback on current project progress
To clarify requirements related to:

- Reflective diary
- Report structure and formatting
- Diagram presentation (WBS, Gantt, 4-Fields Map)

To confirm direction before continuing with design, platform selection, and cost analysis

Agenda Items:
- Review of report progress
- Clarification of coursework requirements
- Feedback from seminar tutor
- Planning next steps

Discussion Summary:
- The seminar tutor reviewed the project progress and confirmed that the work was on the right track.
- Team collaboration was noted to be strong and effective, with no concerns raised.
- General feedback was provided on:
    - Maintaining clear report structure and formatting
    - Ensuring proper diagram presentation and clarity
    - Structuring the reflective diary appropriately

The tutor advised the team to continue developing:

- Design and layout
- Platform selection
- Cost analysis and research

Decisions Made:
The team will continue with the current approach as it aligns with project expectations.

- Focus will be placed on:
    - Improving formatting and clarity
    - Developing diagrams (WBS, Gantt, 4-Fields Map) correctly

- No major changes to the overall project plan were required.

Tasks Allocated:

Task	Assigned
Continue development of 4-bedroom house design and improve report formatting	Raghad
Refine WBS and assist with Gantt Chart updates	All team members
Continue elderly home design, contribute to reflective diary and improve report formatting	Zainab
Work on improving diagrams, report clarity, and begin cost analysis research	All members

Progress Review:

Current project status: On track

Comparison with previous plan:

- Completed:
    - Initial designs and concepts
    - PID and planning documents
    - Draft versions of WBS and Gantt Chart
    - Tutor feedback sessions
- Pending:
    - Finalization of diagrams (WBS, Gantt, 4-Fields Map)
    - Detailed cost analysis
    - Completion of digital models
    - Final report compilation

Changes made:

- Minor refinements to report structure and diagram presentation
- Improved organization of project documentation

Reason:

- Based on tutor feedback to enhance clarity, presentation quality, and alignment with coursework requirements

Next Steps:

- *Finalize diagrams and ensure correct formatting*
- *Continue development of digital house designs*
- *Begin detailed cost analysis*
- *Progress further on report writing and reflective diary*

Meeting Conclusion:
The meeting confirmed that the project is progressing well and aligned with expectations. Constructive feedback from the tutor will help improve clarity and presentation, ensuring a higher-quality final submission.

WEEK 3 :

MEETING 1:

Date: 06/04/2026 **Time:** 7:00 pm **Attendees:** Raghad, Rabia, Zainab **Session:** Group

Meeting Objective:
- To review progress of digital designs and layouts
- To continue development of project diagrams (WBS, Gantt, 4-Fields Map)
- To progress report writing and formatting
- To identify any issues before final tutor review

Agenda Items:

- Review of house designs and layouts
- Progress on digital modelling (Planner 5D)
- Update on diagrams and coding (WBS, Gantt, 4-Fields Map)
- Report writing and formatting progress
- Planning final improvements

Discussion Summary:

- All members presented progress on their assigned housing designs, with most layouts nearing completion
- Digital modelling using Planner 5D was significantly advanced, with key features and layouts implemented
- Diagrams including WBS and Gantt Chart were mostly completed, with minor corrections required in formatting and coding
- The 4-Fields Map was reviewed and identified as needing improvement before final submission
- Report writing was in progress, with focus on improving structure, clarity, and consistency
- Minor issues discussed included:
    - Ensuring proper labeling of designs
    - Improving diagram clarity
    - Aligning all sections of the report

Decisions Made:

- Final refinements will be made to all house designs
- Diagrams will be reviewed and corrected before tutor submission
- Report formatting and structure will be improved
- All components will be aligned for consistency

Tasks Allocated:

Task	Assigned
Finalize 4-bedroom house design and report sections	Raghad
Complete 2-bedroom house design	Rabia
Refine elderly home design, update 4-Fields Map and report sections	Zainab
Improve report formatting and labeling	All members
Review and correct diagrams (WBS, Gantt)	All members

Progress Review :

Current project status: On track

Comparison with previous plan:

Completed:

- Majority of house designs and layouts

- Digital modelling progress
- Draft versions of WBS and Gantt Chart
- Ongoing report writing

Pending:

- Final refinements to designs
- Completion of 4-Fields Map
- Final diagram corrections
- Report formatting and references

Changes to Plan:

Changes made:

- Minor adjustments to design layouts and feature placements
- Refinement of diagrams and report formatting

Reason:

- To improve clarity, accuracy, and overall presentation quality

Next Steps:

- Finalize all designs and digital models
- Complete and correct all diagrams
- Improve report formatting and add references
- Prepare for final tutor review

Meeting Conclusion:

The project is progressing well with most components nearing completion. The team is now focused on refining designs, improving diagrams, and ensuring the report meets the required standards before final submission.

MEETING 2:

Date: 09/04/2026 **Time:** 4:00 pm **Attendees:** Raghad, Rabia, Zainab, Mukarram **Session:** Seminar

Meeting Objective:

- To receive feedback on the project report

- To clarify questions related to report structure and content
- To present current project progress to the seminar tutor

Agenda Items:

- Questions to be clarified
- Review of report progress
- Feedback from seminar tutor

Discussion Summary:

- The seminar tutor reviewed the current progress of the report and provided feedback on key areas.
- It was identified that the 4-Fields Map required regeneration due to incorrect formatting.
- Improvements were suggested for report formatting to ensure consistency and professionalism.
- Additional coding and structure updates were required for the WBS and Gantt chart to improve clarity and alignment with project planning standards.

Decisions Made:

- The correct format of the 4-Fields Map was finalised and agreed for regeneration.
- The team agreed to collaboratively complete the WBS and Gantt chart coding updates.
- Formatting improvements were prioritised to enhance the overall quality of the report.

Tasks Allocated:

Task	Assigned
WBS and Gantt chart updates	All team members
4-fields map Regeneration	Zainab
Report formating	All team members
Layout labeling	All team members

- WBS , gantt chart - Rabia , Raghad, Zainab
- 4 fields map - Zainab
- Formatting - Zainab , Raghad, Rabia

Progress Review :

- Current project status: On track
- Comparison with previous plan:
    - Design layout
    - 4 fields map
    - Diagrams coding

Changes to Plan :

- Changes made:
    - Change in the 4 fields map format
    - Updation of design layout with the addition of labels

- Reason:
    - Incorrect format
    - Detailing for the smart home design

Next Steps:

- Complete the 4 fields map
- Format the report
- add references
- Add WBS and Gantt chart codings
- Label the designs

Meeting Conclusion:

WEEK 4 :

MEETING 1:

Date: 11/04/2026 **Time:** 3:55 pm **Attendees:** Raghad, Rabia, Zainab, Mukarram **Session:** Seminar

Meeting Objective:

- To receive feedback on the final report draft
- To clarify the reflective diary format
- To review presentation content and structure

Agenda Items:

- Final report feedback
- Reflective diary clarification
- Presentation structure and content

Discussion Summary:

- The seminar tutor reviewed the final report and confirmed that it is well-structured and progressing well.
- Positive feedback was given on the overall organisation and quality of the report.
- The reflective diary format was reviewed and confirmed to be appropriate.
- Additional improvements were suggested, including:
- Adding a Stakeholder Power Matrix
- Including an Incident Response and Recovery Diagram

Guidance was provided on improving presentation layout and content clarity.

Decisions Made:

The report will be updated to include:

- Stakeholder Power Matrix
- Incident Response and Recovery Diagram
- The current report structure will be maintained, with minor improvements based on feedback

The team will proceed with finalising the presentation content and design

Tasks Allocated:

Task	Assigned
Stakeholder Power matrix	Zainab
Incidental response recovery plan	All team members
Report formating	All team members
Presentation data and layout	All team members

Progress Review:

- Current project status: On track
- Comparison with previous plan:
- Completed tasks:
    - 4-fields map updated
    - Report formatted
    - Designs labeled
    - References added
    - WBS and Gantt chart codings completed

Pending Tasks:

- Stakeholder Power Matrix
- Incident Response and Recovery Diagram
- Final presentation preparation
- Final report refinements

Changes to Plan:

- Changes made:
    - Inclusion of Stakeholder Power Matrix
    - Addition of Incident Response and Recovery Diagram
- Reason:
    - Based on seminar tutor feedback to enhance analysis, security coverage, and overall report quality

Next Steps:

- Complete remaining report components
- Finalise Stakeholder Power Matrix and incident response diagram
- Prepare and refine presentation slides
- Conduct final review of report before submission

Meeting Conclusion:

The meeting concluded with positive feedback from the seminar tutor, confirming that the project is well-developed and on track. Minor additions and refinements were identified to further improve the report and presentation. The team will focus on completing these final elements to ensure a high-quality submission.

MEETING 2:

Date: *20/04/2026*
Time: *08:30 pm*
Attendees: *Raghad, Rabia, Zainab*
Session: *Group meeting*

Meeting Objective:

To finalise the project report, implement tutor feedback improvements, complete presentation slides, and prepare for the Q&A session.

Agenda Items:

- Final review and editing of project report
- Implementation of feedback from previous submission
- Completion of presentation slides
- Q&A preparation and anticipated questions discussion
- Final submission readiness check

Discussion Summary:

The group collaboratively reviewed the entire report and refined sections based on tutor feedback. Improvements were made to structure, clarity, and academic presentation. The presentation slides were developed and aligned with the report to ensure consistency and smooth flow. The team also discussed potential Q&A questions and prepared collective responses to strengthen understanding and delivery.

Decisions Made:

- Final report approved after minor collaborative refinements
- Presentation slides finalised and aligned with report content
- Q&A preparation strategy agreed as a group
- Final rehearsal scheduled before submission/presentation

Tasks Allocated:
All tasks were completed collaboratively by the group, including report finalisation, slide preparation, and Q&A preparation. Responsibilities were shared equally, with all members contributing to discussion, editing, and refinement of all deliverables.

Current project status: Completed phase

Comparison with previous plan:
What is completed:

- Full report finalised with tutor feedback implemented
- Presentation slides completed and aligned with report
- Q&A preparation completed collaboratively

What is pending:

- Final group rehearsal
- Final submission checks

Changes made:

- Minor restructuring of report sections for improved flow and clarity
- Adjustment of slide order to match final report structure
- Addition of focused Q&A preparation session

Reason:
Changes were made to improve consistency between report and slides, enhance clarity, and ensure alignment with tutor feedback for improved academic quality.

Next Steps:

- Conduct final group rehearsal
- Review presentation timing and delivery
- Final submission of report and slides

Meeting Conclusion:
The meeting concluded with all deliverables finalised collaboratively and the group fully prepared for final rehearsal and submission stage.